
TiME: Test-Time Mixture-of-Experts Routing via Asymmetric CO-Optimal Transport for Continual Test-Time Adaptation

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Abstract

Large language models usually face continuous domain shifts during testing, which degrade performance on unseen shifting domains. So, researchers propose continual test-time adaptation (CTTA) to adapt to evolving testing domains while preserving knowledge of previous domains, making adaptability-stability (A-S) balance. Existing CTTA methods are constrained by dense base models that encode knowledge from all domains into a global model, hardly achieving the A-S balance. We observe that the model sparsity of mixture-of-experts (MoE) models is better for achieving A-S balance than dense models. In CTTA, however, MoE faces difficulty in (1) correctly routing samples from unseen shifting domains and (2) capturing domain-level shifts. In this paper, we propose test-time mixture-of-experts routing (TiME) via asymmetric co-optimal transport (As-COOT): we model MoE routing in CTTA as a test-time allocation problem via COOT. To ensure reliable routing, we propose a semantic space alignment to align sample-expert distributions via bidirectional contrastive learning. To address COOT’s limitations in CTTA, we propose As-COOT, relaxing sample-side constraints while enforcing expert-side constraints to ensure noise robustness and balance expert load. Experiments show TiME outperforms baselines.

1. Introduction

Large language models (LLMs) have been widely applied, and perform well when the testing and source (i.e., training)

domains are aligned. However, in the real world, continuous domain shifts occur during testing, leading to performance degradation when models encounter these unseen domains during testing (Akyürek et al., 2025). For instance, models pre-trained on general tasks often suffer performance drops when continuously applied to unseen specialized tasks (e.g., medicine or law). Therefore, researchers propose the continual test-time adaptation (CTTA) setting (Wang et al., 2022) to enable a source pre-trained model to continuously update parameters to learn from the testing domain (**adaptability**). Meanwhile, the model needs to retain parameters to preserve knowledge from previous domains to prevent catastrophic forgetting (**stability**). The two above are contradictory, so finding the balance between adaptability and stability (**A-S balance**) is essential (Wang et al., 2024a).

To achieve A-S balance, existing CTTA strategies mainly fall into (Wang et al., 2025a): self-regularization (SR) and teacher-student consistency regularization (CR). The SR (Niu et al., 2022; Liu et al., 2024b) introduces regularization to constrain the model updates, preventing extremely far away from the source domain. Although this strategy preserves source-domain knowledge and partially improves stability in CTTA, the constraint restricts the degree of adaptation to the test domain, thereby constraining the adaptability. The CR (Wang et al., 2022; Rafi et al., 2025) uses a student that adapts to the current domain to improve adaptability, while a teacher aggregates past domain knowledge to provide stable guidance, improving stability. However, since the teacher is continuously updated by the student, the teacher tends to be increasingly influenced by recent domains. As domain shifts accumulate, the teacher gradually forgets early domains, limiting long-term stable guidance in CTTA. Both strategies are limited by the model architecture, as it encodes knowledge from all domains into a single global set of parameters, which is typically called *Dense* architectures¹. As a result, adapting the dense models to a new domain inevitably covers parameters that encode knowledge from previous domains, making it challenging for dense models to achieve A-S balance in CTTA.

The *Mixture of Experts* (MoE) architecture (Fedus et al.,

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¹Dense architectures, such as GPT-3, GPT-3.5, are different from MoE architectures (e.g., Qwen1.5-MoE, DeepSeek-V2)

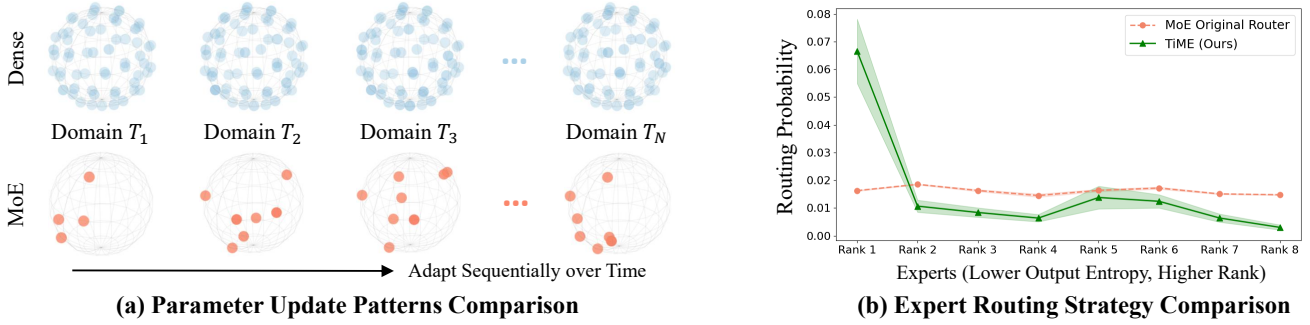


Figure 1. Experimental observations on the limitations of existing architectures under CTTA. (a) **Dense** update all parameters globally at each step (all points are lit). In contrast, **MoE** activates a subset of parameters, adapting locally. (b) An effective routing strategy should allocate samples to suitable experts (e.g., those with low prediction entropy). Under CTTA, the **original MoE router** suffers from routing errors, yielding nearly random expert selection, whereas **TiME (ours)** consistently routes samples to the more suitable (rank-1) expert.

2022) replaces the dense layer of the model with multiple independent expert networks, enabling the use of its sparsity to achieve A-S balance in CTTA. Specifically, MoE relies on a router to allocate samples to different experts, allowing each expert to focus on different domain samples and enabling the model to adapt to the current domain by updating only a few experts without affecting global knowledge. At the same time, experts who retain knowledge from previous domains are kept isolated, avoiding historical domain knowledge being overwritten by continuous parameter updates under CTTA, ensuring stability. Fig. 1a shows the comparison of parameter update patterns between dense and MoE architectures. App. A also shows that MoE architectures, when used as base models during CTTA, outperform dense.

However, despite the above advantages, integrating MoE architectures with CTTA is underexplored and faces challenges as follows: (1) The effectiveness of the router relies on knowledge learned from the source domains during training. When it encounters samples from unseen domains that continuously shifting during testing, the router may allocate them to incorrect experts, leading to routing errors. (2) MoE routing relies on token-level representations, whereas CTTA exhibits continuous domain-level shifts across samples (i.e., shifts that are encoded in the evolving distribution of sample-level statistics). This level mismatch prevents the router from capturing evolving domain characteristics, further leading to routing errors. Fig. 1b shows the routing errors of the original MoE router, where it tends to allocate samples to experts randomly. As a result, routing errors break the A-S balance originally achievable by the MoE.

To mitigate routing errors in CTTA, we argue that it is necessary to explore a CTTA method under the MoE architecture that computes the effective samples-experts allocation scheme at test time based on the current sample and expert parameters, achieving a better A-S balance during CTTA.

In this paper, we propose a test-time mixture-of-experts rout-

ing (**TiME**)² framework via asymmetric co-optimal transport (As-COOT) for continual test-time adaptation (CTTA). We use co-optimal transport (COOT) to model the test-time sample-expert allocation problem. To ensure the effectiveness of the allocation, we align the semantic spaces of samples and experts. To address the limitations of COOT in CTTA with MoE models, we propose As-COOT, which meets the requirements of discarding high-cost noisy samples and constraining expert load balance. Specifically, (1) to enable MoE routing in CTTA, we model it as a test-time sample-expert allocation problem via COOT, which optimizes the minimum-cost allocation scheme from the sample-expert allocation cost, guiding online routing. (2) For the allocation scheme to be semantically meaningful, the sample-expert allocation cost needs to reflect the functional compatibility between samples and experts. To meet this, we propose a sample-expert semantic space alignment module that aligns their representations via bidirectional contrastive learning, so that a distance-based metric can serve as a proxy for the allocation cost, enabling effective cost computation. (3) To mitigate noise effects and achieve balanced expert load, thereby better addressing CTTA, we propose the As-COOT that applies relaxation to the sample-side mass constraint to discard high-cost noisy samples, while imposing expert-side constraints to enforce expert load balance, and we solve As-COOT using block coordinate descent with entropy regularization, which yields a strictly convex objective and allows efficient parallel optimization via the Fenchel-Rockafellar dual and Sinkhorn scaling. TiME outperforms all baselines on the CTTA benchmark, including text understanding and medical reasoning.

Our contributions are: 1) Our analysis substantiates that MoE architecture models are more effective than dense models for CTTA. 2) We model and formalize the test-time MoE routing problem via COOT, and explore the limitations of COOT under CTTA. 3) We propose TiME, a CTTA method

²Code is <https://github.com/coyil-hub/TiME>

for MoE models that effectively mitigates test-time routing errors by using our proposed As-COOT, which relaxes the sample side and constrains the expert side to discard high-cost noise and balance expert loads during CTТА. 4) Experiments show that TiME achieves SOTA performance on the CTТА benchmarks across multiple MoE models.

2. Related Work

Routing in Mixture-of-Experts LLMs. MoE routing allocates input tokens to a subset of experts. It falls into two types (Cai et al., 2025): 1) Training-based routing adopts a parameterized router to map token representations to expert weights. Qwen1.5-MoE (Team, 2024) stabilizes training via load balancing and regularization. DeepSeek-V2 (Liu et al., 2024a) improves expert specialization by fine-grained segmentation. R2-T2 (Li et al., 2025) re-routes guided by reference samples. 2) Training-free routing uses algorithmic allocation to route the expert. Hash Layer (Roller et al., 2021) uses a random router by hashing. THOR (Zuo et al., 2022) randomly routes experts with consistency. Ban&Pick (Chen et al., 2025c) prunes redundant activations. Rewiring (Su et al., 2025) self-supervised via router logit. Training-based routing is tightly coupled to the training domain and thus fails during testing due to domain shift. Training-free routing lacks parameter semantic modeling. We aim to design an online routing to address both limitations.

Continual Test-Time Adaptation. CTТА is a testing process that adapts to continual domain shift (Wang et al., 2022). It falls into: 1) Non-backpropagation-based methods update inputs or outputs. LAME (Boudiaf et al., 2022) adjusts probability with Laplace maximum likelihood. FOA (Niu et al., 2024) uses prediction covariance to learn prompts. 2) Backpropagation-based methods update the model by optimizing. Tent (Wang et al., 2021) adapts by minimizing entropy. CoTTA (Wang et al., 2022) uses a teacher-student framework and data augmentation. SAR (Niu et al., 2023) uses sharpness-aware minimization (SAM). SoTTA (Gong et al., 2023) buffers class-balanced samples and SAM. DeYo (Lee et al., 2024b) adapts images using consistency under perturbation. DSS (Wang et al., 2024b) filters by confidence for positive and negative learning. REM (Han et al., 2025) builds the prediction difficulty via masking. CTТА-T (Liu et al., 2025) perceives domain shift and refines noise by prediction consistency. Existing work focuses on the dense model. Our goal is to explore a CTТА method for Mixture-of-Experts (MoE), as its sparsity is more suitable for CTТА than dense. See more details of related work in the App. P.

3. Method

Overview. To address the routing errors of MoE models in CTТА, we propose TiME (as shown in Fig. 2). (1) The

test-time MoE routing framework (§ 3.1) remodels test-time MoE routing in CTТА as a test-time sample–expert allocation problem. which is instantiated by optimizing a sample–expert allocation scheme via COOT. (2) The semantic space alignment (SSA) module (§ 3.2) aligns the semantic spaces of sample and expert distributions by pushing functionally compatible sample-expert pairs closer via bidirectional contrastive learning, enabling a distance-based sample–expert allocation cost. (3) The As-COOT (§ 3.3) is designed to address the limitations of COOT under CTТА in MoE models by relaxing sample-side constraints and enforcing expert-side constraints, thereby guaranteeing noise discard and balancing expert load. TiME’s pipeline operates at test time, as follows: we first model the test-time MoE routing problem via COOT (§ 3.1). Then, given the current domain sample and expert distributions, TiME aligns their semantic spaces via SSA (§ 3.2). Based on the aligned distributions, As-COOT computes the test-time sample–expert allocation scheme to calibrate the original MoE routing. Finally, we adapt the model with the calibrated experts (§ 3.3).

Problem Formulation. We study CTТА under a MoE architecture, where both the testing domain and expert parameters evolve over time. Model adaptation is conducted online over discrete time steps $t = 1, \dots, T$. At each time step t , the model observes unlabeled test samples $X_t^o = \{x_i^{(t)}\}_{i=1}^N \in \mathbb{R}^{N \times d_s} \sim \mathcal{P}_t$, where $\mathcal{P}_t$ denotes the testing domain distribution. Meanwhile, the MoE model maintains a set of MoE experts, $E_t^o = \{e_j^{(t)}\}_{j=1}^M \in \mathbb{R}^{M \times d_e} \sim \mathcal{D}_t$, whose parameters are continuously updated to adapt to the current domain, and $\mathcal{D}_t$ denotes the expert parameter distribution at time t .

3.1. Modeling Test-Time Routing Problem via COOT

To address the test-time MoE routing (i.e., allocation) problem, CO-Optimal Transport (COOT) (Titouan et al., 2020) provides a paradigm for allocation. We propose a test-time MoE routing (TTR) framework that determines sample–expert allocation schemes at test time based on the current sample and expert distributions, mitigating the original MoE routing errors in CTТА. In MoE, routing schemes are produced by a pretrained router. However, under CTТА, the router, trained on the source domain, may not encounter samples from shifting domains ($\mathcal{P}_t \neq \mathcal{P}_{t-1}$), and may therefore allocate them to inappropriate experts. Consequently, routing should be recomputed online to adapt to evolving test domains rather than relying on the fixed pretrained router. Accordingly, we remodel test-time MoE routing in CTТА as a test-time sample-expert allocation problem.

COOT is a framework for modeling allocation relations between two sets $\mathcal{X}$ and $\mathcal{Y}$. Here, $\mathcal{X} = \{x_i\}_{i=1}^N \in \mathbb{R}^{N \times d_x}$ and $\mathcal{Y} = \{y_j\}_{j=1}^M \in \mathbb{R}^{M \times d_y}$, and COOT models their relations by minimizing an allocation cost $L(\mathcal{X}, \mathcal{Y})$. To enable more stable allocation, COOT jointly optimizes two cou-

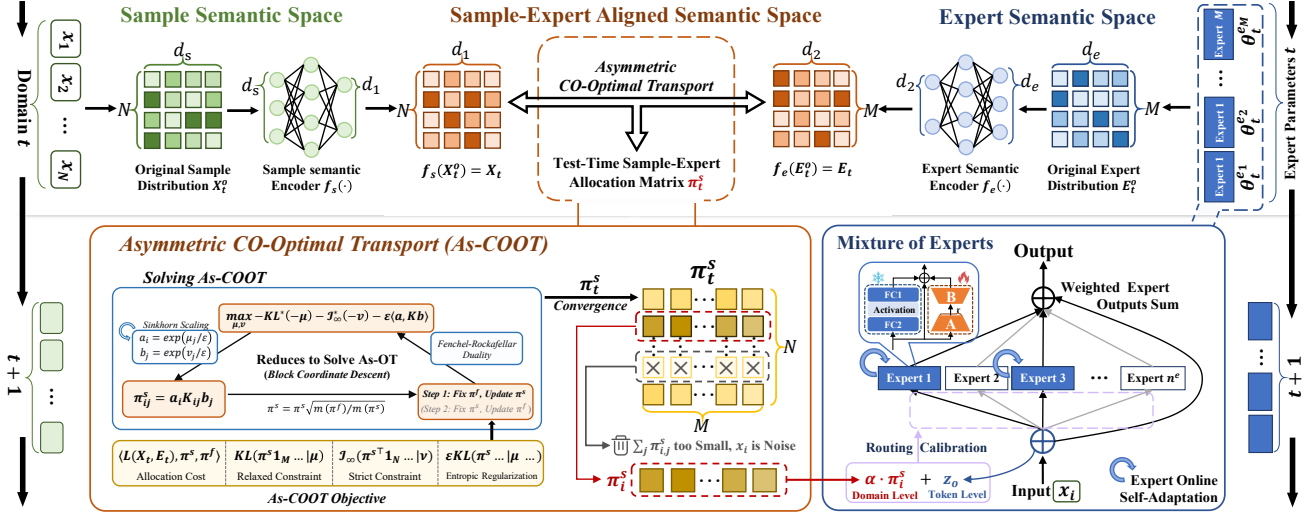


Figure 2. Overview of our proposed TiME. At time step t , firstly, we use a **sample encoder** f_s and an **expert encoder** f_e to encode the current **samples** and the **expert distribution**, X_t^o and E_t^o , into X_t and E_t , respectively, aligning their semantic spaces. Based on X_t and E_t , **As-COOT** is then applied to test-time routing by computing the sample-expert allocation matrix π_t^s during testing, which is used to calibrate the original token-level MoE routing. Finally, we adapt the model online using the experts selected by calibrated routing.

pled matrices: an element allocation matrix $\pi^s \in \mathbb{R}_+^{N \times M}$, where π_{ij}^s represents the allocation weight between element x_i and y_j , and a feature alignment matrix $\pi^f \in \mathbb{R}_+^{d_x \times d_y}$, which adaptively aligns feature dimensions to stabilize the element-level allocation. Both π^s and π^f satisfy symmetric mass conservation, ensuring that each element's total allocation weight matches its prior importance. Specifically, for π^s , mass conservation is imposed over element dimensions with feasible set $\Pi^s(\mu^s, \nu^s) = \{\pi^s \geq 0 \mid \sum_j \pi_{ij}^s = \mu_i^s, \sum_i \pi_{ij}^s = \nu_j^s\}$, where μ^s and ν^s denote prior importance distributions, typically set to uniform: $\mu^s = \frac{1}{N}\mathbf{1}_N$ and $\nu^s = \frac{1}{M}\mathbf{1}_M$. Symmetrically, π^f is defined over a set $\Pi^f(\mu^f, \nu^f)$ with mass conservation imposed on feature dimensions. Accordingly, COOT is a coupled optimization problem that solves for π^s and π^f simultaneously (as Eq. 1).

$$\min_{\pi^s \in \Pi^s, \pi^f \in \Pi^f} L(\mathcal{X}, \mathcal{Y}, \pi^s, \pi^f). \quad (1)$$

Instantiating Test-Time Allocation via COOT. At time step t , we treat the distribution of the current testing samples X_t^o as $\mathcal{X}$, which captures domain characteristics, and expert parameter distribution E_t^o as $\mathcal{Y}$, which encodes the experts' functionality. Based on X_t^o and E_t^o , we optimize the COOT objective (as Eq. 2) to obtain the allocation matrix π_t^s , where $\pi_t^s(i, j)$ is the routing weight from sample $x_i^{(t)}$ to expert $e_j^{(t)}$.

$$\min_{\pi_t^s \in \Pi^s, \pi_t^f \in \Pi^f} L(X_t^o, E_t^o, \pi_t^s, \pi_t^f). \quad (2)$$

Overall, COOT computes a sample-expert allocation matrix π_t^s based on the current sample and expert distributions (X_t^o and E_t^o) for MoE routing during testing in CTTA, which partially mitigates the routing errors of pretrained routers

caused by continuously evolving domains and enables the routing scheme to adapt online to domain-level shifts.

3.2. Semantic Space Alignment for Allocation Cost

To ensure the effectiveness of COOT-based sample-expert allocation, the allocation cost should faithfully reflect the sample-expert functional compatibility. We therefore introduce a sample-expert semantic space alignment (SSA) module, which aligns the semantic spaces of sample and expert via bidirectional contrastive learning. It enables the use of a distance-based metric to model sample-expert functional compatibility as allocation cost.

SSA is crucial, as the effectiveness of COOT for test-time MoE routing critically depends on the quality and interpretability of the allocation cost $L(X_t^o, E_t^o)$ (Eq. 2). However, X_t^o and E_t^o lie in heterogeneous spaces³, rendering the $L(X_t^o, E_t^o)$, computed from original representations, physically meaningless. Our SSA addresses this issue by aligning sample-expert semantic spaces through modeling sample-expert functional compatibility, enabling a distance metric as a proxy for the sample-expert allocation cost. Specifically, we train a sample encoder $f_s(\cdot)$ and an expert encoder $f_e(\cdot)$ to minimize the distance between a sample representation $f_s(x)$ and its compatible expert representation $f_e(e)$ via bidirectional contrastive learning, achieving allocation costs measured by distance metric (lower distance, smaller cost). We train SSA in an offline preprocessing stage, and adopt Euclidean distance $d(x, e) = \|f_s(x) - f_e(e)\|_2$ as the

³ X_t^o captures domain-level sample semantics, whereas E_t^o , the expert parameter distributions, encodes model-level expert functionality, therefore they lie in heterogeneous semantic spaces.

distance metric.

Step 1: Sample-Expert Pair Construction⁴. We investigate two strategies for constructing sample-expert positive (i.e., compatible) and negative pairs: **(1) Expert Output Entropy (EOE).** Experts tend to produce lower entropy (more confident) predictions on samples they handle well; thus, lower-entropy experts are treated as more compatible for the sample. For input x_i , we compute the output entropy $\mathcal{H}_i$ for all experts. The top- K experts with the lowest entropy form the positive set $\mathbb{P}_i = \{e_j^+ | j \in \text{Top-}K(-\mathcal{H}_i)\}$, the rest experts form the negative set $\mathbb{N}_i$. **(2) Original MoE Routing (OR).** The original MoE router activates experts expected to be functionally compatible with the input sample; thus, activated experts are treated as more compatible. For input x_i , we forward it through the frozen MoE model and define the top- K experts activated by the router R_φ as the $\mathbb{P}_i = \{e_j^+ | j \in \text{Top-}K(R_\varphi(x_i))\}$, the rest form the $\mathbb{N}_i$.

Step 2: Alignment via Bidirectional Contrastive Learning. Given the positive and negative sample-expert pairs, we adopt bidirectional contrastive learning (as Eq. 4) with Euclidean distance to achieve stable sample-expert semantic alignment, encouraging functionally compatible pairs to be closer in aligned semantic space. Specifically, we define a sample-to-expert contrastive loss $\mathcal{L}_{s \rightarrow e}$, aiming to minimize the distance between a sample and its positive experts:

$$\mathcal{L}_{s \rightarrow e} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\sum_{e^+ \in \mathbb{P}_i} \exp(-d(f_s(x_i), f_e(e^+))/\tau)}{\sum_{e \in \mathbb{P}_i \cup \mathbb{N}_i} \exp(-d(f_s(x_i), f_e(e))/\tau)}, \quad (3)$$

where τ denotes the temperature. Symmetrically, we define a reverse contrastive loss $\mathcal{L}_{e \rightarrow s}$ to encourage experts focus on semantically compatible samples in the aligned space, enabling more robust learning. The overall objective is:

$$\mathcal{L}_{\text{SSA}} = \mathcal{L}_{s \rightarrow e} + \mathcal{L}_{e \rightarrow s}. \quad (4)$$

By minimizing $\mathcal{L}_{\text{SSA}}$, we obtain aligned f_s and f_e , such that Euclidean distance in the aligned space directly reflects sample-expert compatibility and serves as allocation cost L .

Usage of SSA. Before performing TTR (§ 3.1), we first encode the original sample and expert distributions into the aligned semantic space using the learned encoders during testing: $X_t = f_s(X_t^o) \in \mathbb{R}^{N \times d_1}$ and $E_t = f_e(E_t^o) \in \mathbb{R}^{M \times d_2}$. We then compute the sample-expert allocation cost online using Euclidean distance (i.e., $L(X_t, E_t) = \|X_t - E_t\|_2$). After using SSA, the allocation matrix π_t^s better reflects sample-expert functional compatibility, enabling reliable and meaningful sample-expert allocation.

⁴In CTTA, the testing domain is unseen during the offline stage. All sample-expert pairs use a domain-agnostic C4 corpus (Raffel et al., 2020), following prior work (Chen et al., 2025a).

3.3. Test-Time Routing and Adaptation via As-COOT

To address the limitation imposed by COOT’s symmetric mass conservation constraint in CTTA, we propose Asymmetric CO-Optimal Transport (As-COOT), which relaxes the sample-side mass constraint while keeping the expert-side mass constraint. It enables the discarding of high-cost noisy samples and enforcing balanced expert loads, resulting in an online MoE routing that better fits the CTTA scenario.

1) Why As-COOT? Although COOT supports basic online MoE routing, its strict symmetric mass conservation constraint makes it incompatible with CTTA. Specifically, COOT requires the sample and expert marginals to strictly match: $\sum_j \pi_{i,j}^s = 1/N$, enforcing that all samples allocate experts with equal weights, and $\sum_i \pi_{i,j}^s = 1/M$, enforcing that all experts receive samples with equal weights. Those strict symmetric constraints are harmful in CTTA: (1) As the domain continuously shifts, it inevitably contains many noisy samples with high allocation costs (Feng et al., 2023). To minimize the global allocation cost, COOT is forced to allocate these high-cost noisy samples, which compromises the routing accuracy of clean samples due to the strict sample-side constraint and leads to incorrect routing. Beyond routing errors, the noisy samples will further contaminate expert parameters during online model adaptation, enabling performance degradation. (2) To encourage more expert participation in the CTTA process to maintain a better A-S balance, it is necessary to keep imposing expert-side constraints to improve expert load balancing.

2) As-COOT Objective. To address the above issue, we propose As-COOT, which constructs an asymmetric mass constraint by relaxing the sample-side constraint via KL divergence to discard high-cost noisy samples, while enforcing a strict expert-side constraint via the indicator function $\mathcal{I}_\infty$ to balance experts’ load. The As-COOT objective is:

$$\min_{\pi_t^s, \pi_t^e} \underbrace{\langle L(X_t, E_t), \pi_t^s, \pi_t^e \rangle}_{\text{Allocation Cost}} + \underbrace{\lambda KL(\pi_t^s 1_M \otimes \pi_t^f 1_{d_2} | \mu)}_{\text{Sample Side: Relaxed Constraint}} + \underbrace{\mathcal{I}_\infty(\pi_t^{s\top} 1_N \otimes \pi_t^{f\top} 1_{d_1} | \nu)}_{\text{Expert Side: Strict Constraint}} + \underbrace{\varepsilon KL(\pi_t^s \otimes \pi_t^f | \mu \otimes \nu)}_{\text{Entropic Regularization}}. \quad (5)$$

Here, $\mathcal{I}_\infty(h | \nu) = \begin{cases} 0, & \text{if } h = \nu, \\ +\infty, & \text{if } h \neq \nu, \end{cases}$, $\mu = \mu^s \otimes \mu^f$, $\nu = \nu^s \otimes \nu^f$, and λ controls the relaxation degree. The entropic regularization term transforms the subproblem (as Eq. 6) into a strictly convex problem (analyses are in App. C), which enables a global optimal solution (see proof of optimality and convergence of As-COOT in App. D). We relax the sample-side marginal constraint via the KL divergence, allowing allocation mass destruction (i.e., $\sum_j \pi_{i,j}^s < 1/N$). For high-cost noisy samples that cannot be reliably allocated to any expert, As-COOT can discard them at a penalty λ , which endows the routing with noise robustness in CTTA (see analyses of noise in App. F). Asymmetrically, we use

$\mathcal{I}_\infty$ to enforce that the expert-side marginal strictly matches the prior distribution ν (i.e., $\sum_i \pi_{i,j}^s = 1/M$), which guarantees mass conservation to ensure sufficient sample allocation to each expert, achieving better balance expert loads.

3) Solving As-COOT. The As-COOT objective is non-convex in (π_t^s, π_t^f) , so we solve it by block coordinate descent (BCD) (Titouan et al., 2020), alternately solving its two nested Asymmetric OT (As-OT) subproblems (as Eq. 6, see that As-COOT is simplified into As-OT in the App. E). **Step 1: Fix π_t^f , update π_t^s .** The objective of As-OT is:

$$\min_{\pi_t^s} \langle C, \pi_t^s \rangle + \lambda \text{KL}(\pi_t^s \mathbf{1}_N | \mu^s) + \mathcal{I}_\infty(\pi_t^s \mathbf{1}_{d_1} | \nu^s) + \varepsilon \text{KL}(\pi_t^s || \mathbf{1}), \quad (6)$$

where $C = L(X_t, E_t, \pi_t^f)$. To facilitate subsequent numerical computation (i.e., Sinkhorn scaling (Cuturi, 2013)), we apply the Fenchel-Rockafellar dual (Rockafellar, 1967) to convert the subproblem into (detailed dual see App. B.2):

$$\max_{\phi, \psi} -\text{KL}^*(-\phi) - \mathcal{I}_\infty^*(-\psi) - \varepsilon \sum_{i,j} \exp\left(\frac{\phi_i + \psi_j - C_{ij}}{\varepsilon}\right), \quad (7)$$

where ϕ and ψ are dual variables. We then solve Eq. 7 via Sinkhorn scaling. We define the sample-side and expert-side scaling factors as $a_i = \exp(\phi_i/\varepsilon)$ and $b_j = \exp(\psi_j/\varepsilon)$, respectively, and the kernel as $K_{ij} = \exp(-C_{ij}/\varepsilon)$. We alternately update a_i and b_j until they converge, and finally recover the allocation matrix entry as $\pi_{ij}^s = a_i K_{ij} b_j$.

Step 2: Fix π_t^s , update π_t^f . This step is analogous to Step 1. We provide the detailed solution of As-OT (i.e., Step 1 or 2) in App. B. After each step, we perform rescaling $\pi_t^{s/f} = \pi_t^{s/f} \sqrt{m(\pi_t^{s/f})/m(\pi_t^{f/s})}$, where $m(\pi) = \sum_{i,j} \pi_{ij}$, to remove the scaling uncertainty. The optimization terminates upon convergence of π_t^s and π_t^f . We obtain the current sample-expert allocation matrix π_t^s by solving As-COOT.

4) Routing Calibration at Test Time via As-COOT. Solving As-COOT, we obtain the allocation matrix π_t^s , which characterizes the current sample-expert allocation. However, this domain-level π_t^s cannot be directly applied to the token-level routing z_o of the original MoE router R_φ . To address this, we propose a routing calibration strategy that uses π_t^s to calibrate the z_o during testing. Specifically, we row-normalize π_t^s to obtain $\widetilde{\pi}_t^s$, which serves as a domain-level prior over allocation under the current domain. We then define the calibrated routing distribution z_{As} as Eq. 8 (detailed explanation of z_{As} is in App. H.2).

$$z_{As} = z_o + \alpha \cdot \log(\widetilde{\pi}_t^s + \epsilon), \quad (8)$$

where α controls calibration strength and a small constant ϵ prevents $\log(0)$. By using the current allocation scheme $\widetilde{\pi}_t^s$ to calibrate the original routing z_o , the resulting z_{As} jointly captures token-level semantic information and domain-level allocation computed online by As-COOT, mitigating routing errors of the pretrained router during testing in CTTA.

5) Adaptation via As-COOT-based Calibrated Routing.

After routing calibration, the model obtains a more reliable routing z_{As} . Based on z_{As} , we perform online adaptation by minimizing model prediction uncertainty under the current domain. Specifically, we first define the prediction distribution: given the context $\mathcal{H}$, we aggregate next-token w predictions $p_e(w | \mathcal{H})$ from the top- K experts $e \in \mathcal{Z}_{\text{top-k}}$ selected by z_{As} , weighted by their routing scores $z_{As}^{(e)}$, to form the final prediction $p(w | \mathcal{H}) = \sum_{e \in \mathcal{Z}_{\text{top-k}}} z_{As}^{(e)} p_e(w | \mathcal{H})$. Using this calibrated prediction, we update the model by minimizing the prediction entropy (as Eq. 9).

$$\mathcal{L}_{\text{ent}} = - \sum_w p(w | \mathcal{H}) \log p(w | \mathcal{H}). \quad (9)$$

Minimizing $\mathcal{L}_{\text{ent}}$ encourages more confident predictions under the evolving domain. Model parameters are updated from θ_t to θ_{t+1} ⁵. However, optimizing solely with $\mathcal{L}_{\text{ent}}$ may lead the model to overfit the current domain and partially forget source-domain knowledge, incurring catastrophic forgetting. To mitigate this, we randomly restore partial θ_t to initial states θ_0 , thereby injecting source-domain knowledge. Specifically, for each parameter tensor, we sample a binary mask $M \in \{0, 1\}^d$, $M \sim U(p)$, where p is the restore probability; masked parameters are reset to θ_0 (as Eq. 10).

$$\theta_{t+1} = M \odot \theta_0 + (1 - M) \odot \theta_{t+1} \quad (10)$$

$\mathcal{L}_{\text{ent}}$ aims to adapt the model to current domain knowledge, while the random restoration prevents excessive parameter drift. Together, they ensure the model adapts quickly to the testing domain while preserving source domain knowledge.

4. Experiments

Benchmark and Datasets. Based on prior work (Liu et al., 2025; Chen et al., 2025a), we construct the CTTA benchmark, which includes two task streams: text understanding (**TU**) and medical reasoning (**Med**). (1) The TU stream includes the Winogrande (Sakaguchi et al., 2021), the Easy and Challenge sets in the ARC (Clark et al., 2018), BoolQ (Clark et al., 2019), OpenBookQA (Mihaylov et al., 2018), RTE (Bentivogli et al., 2009), and the Chinese and Spanish sets in XQuAD (Dumitrescu et al., 2021). (2) The Med stream includes the ICD9-CM, ICD9-PROC, ATC, ICD10-CM, and ICD10-PROC sets in MedConceptsQA (Shoham & Rappoport, 2024), MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), and PubMedQA (Jin et al., 2019). The model sequentially adapts to datasets within each task stream in order during testing, following the CTTA setting (Wang et al., 2022; Han et al., 2025; Liu et al., 2025).

Baselines. We compare our TiME with mainstream CTTA methods. *Tent* (Wang et al., 2021), *CoTTA* (Wang et al.,

⁵We inject LoRA adapters (Hu et al., 2022) into all experts and update only the adapters θ_t while keeping the base model frozen.

Table 1. Comparison of different CTТА methods on the Text Understanding task stream across two MoE models. **TiME (EOE)** uses expert output entropy in SSA (§ 3.2), and **TiME (OR)** uses the original MoE routing. **Bold** and Underline denote the best and second-best. Source denotes zero-shot predictions without adaptation. All results are statistically significant under t-test ($p < 0.01$, details in App. Q).

Base Model	Method	t								Avg $\uparrow$
		Winogrande Acc	ARC-c Acc	ARC-e Acc	BoolQ Acc	OBQA Acc	RTE Acc	XQuAD-Zh EM	XQuAD-Es EM	
Qwen1.5-MoE-A2.7B	Source	0.6531	0.3179	0.6520	0.7204	0.2734	0.6458	0.2111	0.0851	0.4449
	Tent	0.6401	0.3159	0.5642	0.6579	0.2657	0.6657	0.0000	0.0000	0.3887
	CoTTA	0.6498	0.3179	0.5734	0.7064	0.2643	0.6159	0.0496	0.0277	0.4006
	SAR	0.6357	0.2861	0.6562	0.7190	0.2756	0.6592	0.1092	0.0462	0.4234
	SoTTA	0.6480	0.2936	0.6562	0.7197	<u>0.2780</u>	0.6642	0.0992	0.0496	0.4261
	DSS	0.6357	0.3066	0.6404	0.7204	0.2720	0.6710	0.0782	0.0647	0.4236
	REM	0.6380	0.3106	<u>0.6573</u>	0.7204	0.2767	0.6810	0.1042	0.0429	0.4301
	CTTA-T	0.6577	0.3238	0.6498	0.7169	0.2767	0.6956	0.2252	0.0924	0.4548
	TiME (EOE)	<u>0.6687</u>	0.3421	0.6520	0.7318	0.2812	<u>0.7222</u>	<u>0.2328</u>	<u>0.0983</u>	<u>0.4661</u>
	TiME (OR)	0.6734	<u>0.3368</u>	0.6659	<u>0.7261</u>	0.2812	0.7361	0.2437	0.1025	0.4707
DeepSeek-V2-Lite	Source	0.6650	0.3460	0.6913	0.6711	0.2578	0.5833	0.0580	0.1325	0.4256
	Tent	0.6475	0.3333	0.4140	0.6565	0.2240	0.5539	0.0000	0.0000	0.3537
	CoTTA	0.6419	0.3733	0.6923	0.6692	0.2638	0.5845	0.0168	0.0429	0.4106
	SAR	0.6573	0.3866	0.6923	0.7084	0.2756	0.5986	0.0235	0.0588	0.4251
	SoTTA	0.6622	<u>0.3994</u>	0.6958	0.7118	0.2800	0.5914	0.0210	0.0580	0.4275
	DSS	0.6498	0.3466	0.6853	0.7064	0.2701	0.5880	0.0118	0.0244	0.4103
	REM	0.6577	0.3866	0.6853	0.7143	0.2680	0.5942	0.0202	0.0311	0.4197
	CTTA-T	<u>0.6667</u>	0.3933	0.6958	0.7098	<u>0.2812</u>	0.5942	0.0294	0.0697	0.4300
	TiME (EOE)	<u>0.6667</u>	<u>0.3982</u>	0.6993	0.7197	0.2738	<u>0.5971</u>	0.0706	<u>0.1345</u>	<u>0.4450</u>
	TiME (OR)	0.6719	0.4013	<u>0.6979</u>	<u>0.7152</u>	0.2857	0.6052	<u>0.0668</u>	0.1407	0.4481

2022), SAR (Niu et al., 2023), SoTTA (Gong et al., 2023), DSS (Wang et al., 2024b), REM (Han et al., 2025), and CTТА-T (Liu et al., 2025).

Base Models. We conduct the experiments using two base MoE models: Qwen1.5-MoE-A2.7B-Chat (Qwen) (Team, 2024) and DeepSeek-V2-Lite (DeepSeek) (Liu et al., 2024a). The datasets, metrics, baselines, and implementation details are in App. H, and hyperparameter analyses are in App. O.

4.1. Main Results

Main Results of TU Task Stream (as Tab. 1). Our proposed TiME achieves the best average performance across all CTТА baselines on both MoE models. TiME improves the average score by +1.59% over the second-best baseline (CTТА-T) on Qwen and by +1.81% on DeepSeek. Notably, several baselines (e.g., Tent and CoTTA) underperform the source model, showing the difficulty of improving performance under realistic and continuously shifting testing domains. Tent yields 0% on XQuAD, showing that such simple methods lead to severe model collapse in hard generative tasks; a similar phenomenon is also observed in other baselines. Within the SSA module (§ 3.2), the two sample-expert compatibility construction strategies (EOE and OR) achieve similar performance, showing that SSA is insensitive to the specific construction strategy and can stably capture sample-expert compatibility as long as reasonable compatible signals are provided. See comparisons with methods of *online MoE re-routing* setting in App. L.

Main Results of Med Task Stream (as Tab. 4). Our TiME also achieves the best average score on Qwen, showing that our TiME maintains consistent and strong performance even on specialized tasks (i.e., medical reasoning). Consistent with the TU task stream, several CTТА baselines fail to surpass the zero-shot source model, showing the difficulty of reliable adaptation under CTТА in specialized domains.

We conduct (1) **evaluation of time and memory efficiency** (as Tab. 5), (2) **evaluation of the efficiency of each module of TiME** (as Tab. 6), and (3) **analyses of task stream order** (as Tab. 7) in App. I, App. J, and App. K, respectively.

4.2. Ablation Studies

We conduct comprehensive ablation studies on (1) test-time routing strategies (§ 3.1, § 3.3), (2) the distance metric of SSA (§ 3.2), (3) regularization strategies (§ 3.3.5) to validate the necessity of each component, as shown in Tab. 2.

Test-Time Routing Strategy Ablation. Replacing AsCOOT with COOT degrades performance, showing that asymmetric mass constraint is crucial for discarding high-cost noisy samples and achieving expert load balancing during CTТА, thereby mitigating noise pollution and enabling more experts to participate in CTТА. Using Euclidean-distance-based routing leads to a larger drop, as shallow metric-based allocation fails to consider allocation difficulty. Classical training-free routing methods (i.e., Hash Layer that uses a random router by hashing the tokens) perform worse, as they ignore the sample-expert functional compatibility

Table 2. Ablation study of different components. "r." denotes replacement with the corresponding module or strategy. "w/o." denotes removing it entirely. The experiments are conducted on the Text Understanding task stream using Qwen1.5-MoE-A2.7B-Chat.

Method	Winogrande	ARC-c	ARC-e	BoolQ	OBQA	RTE	XQuAD-Zh	XQuAD-Es	Avg $\uparrow$
TiME (ours)	0.6734	0.3368	0.6659	0.7261	0.2812	0.7361	0.2437	0.1025	0.4707
<i>Test-Time Routing (Allocation) Strategy Ablation</i>									
r. COOT-based	0.6656	0.3238	0.6520	0.7184	0.2773	0.6805	0.2272	0.0975	0.4553
r. Euclidean-distance-based	0.6458	0.2975	0.6404	0.6843	0.2734	0.6705	0.2018	0.0950	0.4386
r. Hash Layer (Roller et al., 2021)	0.6371	0.2975	0.6404	0.6791	0.2748	0.6697	0.1874	0.0765	0.4328
w/o. As-COOT	0.6391	0.2947	0.6312	0.6707	0.2692	0.6875	0.1933	0.0977	0.4354
<i>Distance-based Metric of SSA Module Ablation</i>									
r. Cosine distance	0.6696	0.3341	0.6597	0.7261	0.2840	0.7375	0.2395	0.0992	0.4687
w/o. SSA	0.6509	0.3196	0.6381	0.7143	0.2773	0.7152	0.2210	0.0773	0.4517
<i>Regularization Strategy Ablation</i>									
r. L2 regularization	0.6656	0.3245	0.6487	0.7254	0.2812	0.7285	0.2344	0.1042	0.4641
w/o. Random restoration	0.6562	0.3221	0.6451	0.7247	0.2774	0.7081	0.2008	0.0924	0.4534

and rely on heuristic allocation. Removing online routing and reverting to the original routing causes notable degradation, as the pretrained routing learned from the source domain cannot adapt to continuous domain shifts during testing and leads to routing errors.

SSA Distance-based Metric Ablation. We perform studies on the distance metric used in the SSA module § 3.2 to validate its robustness and necessity. Replacing Euclidean with cosine distance causes only minor performance changes, showing SSA’s robustness to the specific choice of distance metric. Removing SSA degrades performance, as missing semantic alignment prevents As-COOT from computing a physically meaningful allocation π^s based on sample-expert compatibility, leading to inferior routing quality.

Regularization Strategy Ablation. Replacing random restoration with L2 regularization (Tikhonov, 1963) causes a slight performance drop, suggesting that preserving source-domain knowledge requires exploring alternative regularization methods. Removing random restoration degrades performance, showing that although the MoE architecture provides a degree of A-S balance, explicitly injecting source-domain knowledge results in more stable adaptation.

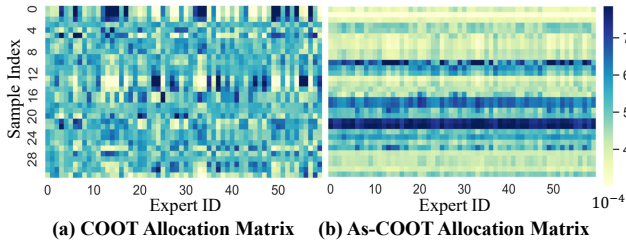


Figure 3. The heatmaps of the allocation matrix π^s of As-COOT and COOT. Rows correspond to samples and columns correspond to experts, with darker colors indicating higher allocation weights. As-COOT shows lighter rows for noisy samples.

4.3. Analysis Studies

Analyses of Noise Discarding of As-COOT. To verify the noise discarding ability of As-COOT under CTTA, we compare the allocation matrices π^s of As-COOT and COOT (as Fig. 3) and visualize them, where darker rows indicate higher allocation preference (i.e., lower allocation cost and clean samples), and lighter rows indicate the opposite (i.e., noisy samples). For COOT, π^s_{COOT} shows uniform colors (as Fig. 3a) because strict mass conservation forces noisy samples, despite their high costs, into the optimization, thereby hindering the clean samples routing. For As-COOT, in contrast, $\pi^s_{\text{As-COOT}}$ shows a clear layered structure (as Fig. 3b). By explicitly relaxing sample-side constraints, As-COOT suppresses the allocation weights of noisy samples (visualized as lighter rows), which mitigates noise accumulation. The theoretical analyses of the noise robustness of As-COOT compared to COOT are in the App. F.

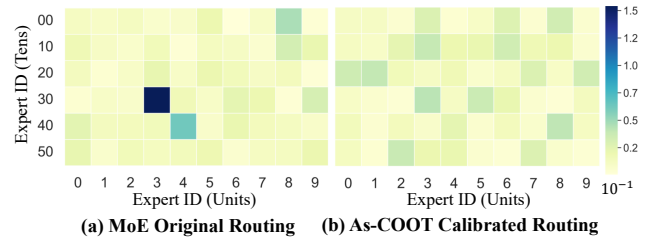


Figure 4. The heatmaps of experts’ activation frequency. The 60 experts are reshaped into a 6×10 grid by expert ID (rows: tens, columns: units). Darker colors indicate higher activation frequency, with As-COOT showing more balanced usage.

Analysis of Expert Load Balancing of As-COOT. To verify that As-COOT improves expert load balancing under CTTA, we compare it with the original MoE routing by visualizing expert activation frequency during CTTA, where darker colors indicate higher usage. Ideally, expert usage

frequency (color) approaches a uniform distribution. As Fig. 4a, the original MoE exhibits clear bias, activating only a few experts frequently while underutilizing others. In contrast, the As-COOT-based calibrated routing (as Fig. 4b) yields a more uniform usage, thanks to its strict expert-side marginal constraint, which guarantees load balancing. The original MoE, relying on source-domain routing patterns, often routes unseen samples to a small subset of experts incorrectly under CTTA, leading to load imbalance.

5. Conclusion

We propose TiME, a novel test-time MoE routing method for CTTA. We model a test-time sample-expert routing problem via COOT, which enables online allocation between current domain samples and MoE experts during CTTA. To ensure effective routing, we align sample and expert distributions in an aligned semantic space, modeling functional compatibility as a distance-based allocation cost. Above all, we propose As-COOT to address the limitations of COOT in CTTA with MoE models by relaxing sample-side constraints and enforcing expert-side constraints, thereby ensuring effective noise discarding and expert load balancing. Experiments show that TiME achieves SOTA performance.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Appendix: TiME: Test-Time Mixture-of-Experts Routing via Asymmetric CO-Optimal Transport for Continual Test-Time Adaptation

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A. Effectiveness and Suitability of Base Model Architecture in CTTA: MoE vs. Dense

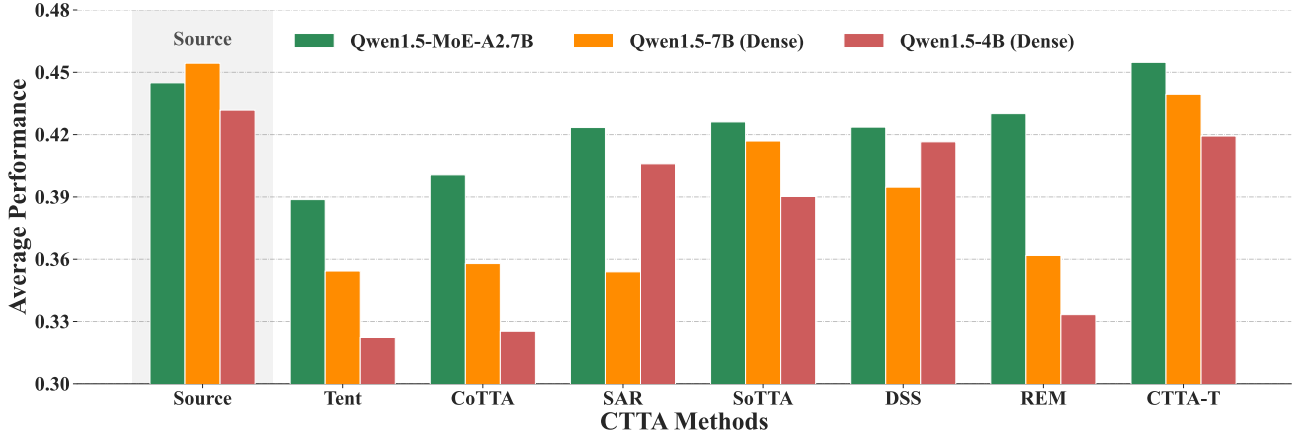


Figure 5. Performance comparison of MoE (Qwen1.5-MoE-A2.7B) and Dense (Qwen1.5-7B and Qwen1.5-4B) architectures when used as base models under various CTTA methods on Text Understanding task stream. Source denotes zero-shot predictions without adaptation.

To empirically validate the structural effectiveness and suitability of MoE over Dense architectures as base models for CTTA, we conducted a comparative experiment using Qwen1.5-MoE-A2.7B against two dense models: Qwen1.5-4B, which has a comparable number of active parameters, and Qwen1.5-7B, which exhibits stronger zero-shot performance. The comparison is conducted across seven mainstream CTTA methods. As shown in Fig. 5, the Dense-7B model achieves the highest zero-shot performance (0.4544), which is expected given its larger parameter scale. However, this advantage does not persist under continuous adaptation. Specifically, under methods like Tent and CoTTA, the performance of the Dense-7B sharply degrades to 0.3543 and 0.3579, respectively, falling below the MoE baseline. In contrast, Qwen1.5-MoE-A2.7B shows substantially better robustness during adaptation. Under Tent, it achieves 0.3887, clearly outperforming both Dense-4B (0.3223) and Dense-7B (0.3543). Moreover, when combined with CTTA-T, the MoE model reaches a peak performance of 0.4548, which even exceeds the original zero-shot performance of the much larger Dense-7B model. Although one might expect Dense-7B to maintain its superiority during adaptation, the results reveal the opposite trend. Dense architectures, when adopted as CTTA base models, are highly vulnerable to continuous adaptation in CTTA settings, whereas the MoE architecture remains significantly more stable.

To further investigate whether MoE architectures provide a better adaptability-stability trade-off in CTTA, we additionally evaluate representative methods using two continual-adaptation metrics: average performance (AP), which measures overall adaptability across the task stream, and average forgetting (AF), which quantifies how much previously learned knowledge is forgotten during adaptation. The comparison is conducted on both Dense and MoE architectures. As shown in Tab. 3,

Table 3. Comparison of Dense and MoE architectures under CTTA from the perspectives of adaptability and stability. AP denotes average performance ($\uparrow$), while AF denotes average forgetting ($\downarrow$).

Model	Tent	REM	CTTA-T	TiME
Qwen1.5-7B (Dense)	0.4169/0.0265	0.3333/0.0165	0.4193/0.0154	-
Qwen1.5-MoE-A2.7B (MoE)	0.4261/0.0169	0.4301/0.0103	0.4548/0.0055	0.4707/0.0027

MoE consistently achieves both higher AP and lower AF than Dense across different CTTA methods. For example, under Tent, the MoE model improves AP from 0.4169 to 0.4261 while reducing AF from 0.0265 to 0.0169. Similar trends can be observed for REM and CTTA-T, where MoE simultaneously enhances adaptation capability and mitigates catastrophic forgetting. Furthermore, TiME on the MoE architecture achieves the best overall result, obtaining the highest AP (0.4707) together with the lowest AF (0.0027).

These results provide evidence that *the MoE architecture is more suitable as a base model for CTTA*. Although the Dense-7B model starts with strong source (zero-shot) performance, it experiences severe performance degradation during continuous adaptation in CTTA. In contrast, the MoE model consistently outperforms both Dense-4B and Dense-7B under most CTTA methods and achieves both higher AP and lower AF, demonstrating a superior balance between adaptability and

stability. This advantage comes from the sparse expert structure of MoE. During adaptation, only a subset of experts needs to update to fit the current domain, which improves adaptability. Meanwhile, the remaining experts stay largely isolated, helping preserve global knowledge and maintain overall stability.

Nevertheless, it is important to note that even when adopting a MoE architecture that is more suitable as a CTTA base model, existing CTTA methods still struggle to fully exploit its potential. As shown in Fig. 5, under most CTTA methods, the adapted performance of the MoE model remains comparable to or even below its zero-shot performance, indicating limited adaptation gains. This observation suggests that the current CTTA algorithms are not sufficiently effective when applied to MoE-based models, and in many cases fail to surpass simple zero-shot inference without adaptation. Therefore, merely replacing the base model with a more suitable MoE architecture is not enough. It is necessary to develop a more effective CTTA method that is specifically designed to align with the characteristics of MoE architectures, in order to unlock their full potential under CTTA.

B. Asymmetric Optimal Transport (As-OT, Sub-problem of As-COOT) Solution Derivation

B.1. As-OT Solution Derivation

We take the optimization of π^s as an illustrative example.

1. Primal Problem Setup

Given a set of samples (with N samples) and a set of experts (with M experts), together with an allocation cost matrix $C \in \mathbb{R}^{N \times M}$, we aim to find an allocation matrix π^s that simultaneously minimizes the allocation cost, satisfies sample-side and expert-side marginal constraints, and includes entropy regularization to ensure numerical solvability and parallelization. We first write the entropy-regularized As-OT primal problem:

$$\min_{\pi^s \in \mathbb{R}_+^{N \times M}} \langle C, \pi^s \rangle + \lambda \text{KL}(\pi^s \mathbf{1}) + \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}) + \varepsilon \text{KL}(\pi^s \parallel \mathbf{1}), \quad (11)$$

where $C \in \mathbb{R}^{N \times M}$ denotes the cost of allocating sample i to expert j (input). $\pi^s \in \mathbb{R}^{N \times M}$ denotes the allocation matrix, where $\pi_{i,j}^s$ is the weight of allocating sample i to expert j (output). $\langle C, \pi^s \rangle = \sum_{i,j} C_{ij} \pi_{ij}^s$ is the optimization objective. Minimizing $\langle C, \pi^s \rangle$ encourages π^s to place mass on entries with smaller cost (i.e., better expert allocations). $\pi^s \mathbf{1} \in \mathbb{R}^N$ is the row-sum vector, where $(\pi^s \mathbf{1})_i = \sum_j \pi_{ij}^s$ is the sum of the i -th row. $\pi^{s\top} \mathbf{1} \in \mathbb{R}^M$ is the column-sum vector, where $(\pi^{s\top} \mathbf{1})_j = \sum_i \pi_{ij}^s$ is the sum of the j -th column. $\text{KL}(\cdot)$ and $\mathcal{I}_\infty(\cdot)$ control the sample-side and expert-side marginals, respectively. $\varepsilon > 0$ is the entropy regularization strength. A larger ε produces smoother and more uniform (soft) allocations, while a smaller ε leads to sparser or harder routing.

Sample-Side Constraint KL . In the sample-side constraint, we impose a soft constraint through the KL divergence. The corresponding variable is the row-sum vector $s = \pi^s \mathbf{1} \in \mathbb{R}^N$, which represents the allocation mass of each sample. This constraint takes a prior distribution μ^s (usually the uniform weight $1/N$) as reference, and its specific form is

$$\lambda \text{KL}(s \mid \mu^s) = \lambda \sum_i \left(s_i \log \frac{s_i}{\mu_i^s} - s_i + \mu_i^s \right). \quad (12)$$

Its physical meaning is to encourage the allocation mass s_i of each sample i to stay close to the prior weight μ_i^s . However, if a sample corresponds to noise (that is, its allocation costs C_{ij} are overall large), the optimization process tends to push s_i toward zero so as to reduce the total allocation cost. In this way, the model achieves a “rejection” effect for that sample.

Expert-Side Constraint $\mathcal{I}_\infty$. In the expert-side constraint, we impose a hard constraint through the indicator function $\mathcal{I}_\infty$. The corresponding variable is the column-sum vector $h = \pi^{s\top} \mathbf{1} \in \mathbb{R}^M$, which represents the amount of mass received by each expert. This constraint takes a prior distribution ν^s (usually the uniform weight $1/M$) as reference, and its specific form is

$$\mathcal{I}_\infty(h) = \begin{cases} 0, & h = \nu^s, \\ +\infty, & \text{otherwise,} \end{cases} \quad (13)$$

which forces the total mass received by each expert to be exactly equal to the prior distribution ν^s . This design ensures coverage in model parameter updates and improves the load balance across experts.

Entropy regularization $\varepsilon \text{KL}(\pi^s \parallel \mathbf{1})$. $\text{KL}(\pi^s \parallel \mathbf{1})$ treats the allocation matrix π^s as a non-negative distribution and measures its relative entropy with respect to the all-ones matrix. This term penalizes the information content of π^s and thus encourages solutions with higher entropy. In optimization, it acts as a strong smoothing regularizer. It prevents π^s from becoming overly sparse (that is, a sample is fully allocated to a single expert), which benefits both numerical optimization and generalization. Entropy regularization turns the original linear-programming OT problem into a strictly convex problem (see App. C for analysis of strict convexity), which enables efficient solutions via Sinkhorn scaling.

2. Dual Derivation and Sinkhorn Iterations.

We introduce dual variables $\phi \in \mathbb{R}^N$ (for the sample side) and $\psi \in \mathbb{R}^M$ (for the expert side). Through the Fenchel–Rockafellar dual transformation, the primal problem turns into the following dual formulation (see App. B.2 for the detailed derivation of the dual):

$$\max_{\mu^s, \nu^s} \left\{ -KL^*(-\phi) - \mathcal{I}_\infty^*(-\psi) - \varepsilon \sum_{i,j} \exp\left(\frac{\phi_i + \psi_j - C_{ij}}{\varepsilon}\right) \right\}. \quad (14)$$

To facilitate the implementation of the Sinkhorn scaling algorithm, we perform a change of variables $a = e^{\phi/\varepsilon} \in \mathbb{R}_{>0}^N$ and $b = e^{\psi/\varepsilon} \in \mathbb{R}_{>0}^M$. We define the Gibbs kernel matrix $K_{ij} = \exp(-C_{ij}/\varepsilon)$. Note that this mapping is element-wise: $a_i = e^{\phi_i/\varepsilon}$ and $b_j = e^{\psi_j/\varepsilon}$.

By rewriting the dual objective in terms of a and b (and expressing the arguments of the convex conjugates back in logarithmic form), we obtain

$$\max_{a>0, b>0} \left\{ -KL^*(-\varepsilon \log a) - \mathcal{I}_\infty^*(-\varepsilon \log b) - \varepsilon \langle a, Kb \rangle \right\}. \quad (15)$$

This formulation is equivalent to the dual problem expressed in variables a and b . The variables a and b are the scaling factors, and they appear directly when reconstructing π^s :

$$\pi^s = \text{diag}(a) K \text{diag}(b). \quad (16)$$

When computing with a and b , we can perform iterative updates either in the log domain or in the standard numerical domain.

From the Dual Formulation to Sinkhorn Scaling Iterations. We apply Block Coordinate Ascent to the dual objective: fix b and optimize a ; then fix a and optimize b . We derive these two steps separately and give closed-form updates.

Step 1: Fix b and solve the subproblem over a (sample-side update). We compute the sample-side scaling factor a , handling asymmetric constraints and allowing noisy samples to be only partially allocated.

The subproblem is

$$\max_{a>0} -KL^*(-\varepsilon \log a) - \varepsilon \langle a, Kb \rangle. \quad (17)$$

On the sample side, we use a KL soft constraint: $\text{KL}(s) = \lambda \text{KL}(s \mid \mu^s)$. It is most intuitive to solve this via the Proximal View using the marginal variable $s = \pi^s \mathbf{1}$. When b is fixed, the current predicted row sum is target $:= Kb$ (i.e., $\sum_j K_{ij} b_j$). We need to find a new row sum s that stays close to the prior μ^s (controlled by λ) and also close to the predicted value target (controlled by ε). The Prox problem is

$$\min_{s \geq 0} \lambda \text{KL}(s \mid \mu^s) + \varepsilon \text{KL}(s \mid \text{target}). \quad (18)$$

Expanding the objective (dropping constant terms), we obtain

$$\sum_i \left(\lambda s_i \log \frac{s_i}{\mu_i^s} + \varepsilon s_i \log \frac{s_i}{\text{target}_i} \right). \quad (19)$$

Taking the derivative w.r.t. s_i and setting it to zero gives

$$\lambda (\log s_i - \log \mu_i^s + 1) + \varepsilon (\log s_i - \log \text{target}_i + 1) = 0. \quad (20)$$

Rearranging (the constant +1 terms cancel),

$$(\lambda + \varepsilon) \log s_i = \lambda \log \mu_i^s + \varepsilon \log \text{target}_i. \quad (21)$$

Exponentiating yields the optimal marginal

$$s_i = \mu_i^s \frac{\lambda}{\lambda+\varepsilon} \cdot \text{target}_i^{\frac{\varepsilon}{\lambda+\varepsilon}}. \quad (22)$$

By the definition of scaling, the actual row sum satisfies $s = a \odot (Kb) = a \odot \text{target}$. Hence $a_i = s_i / \text{target}_i$. Substituting the expression above,

$$a_i = \frac{\mu_i^s \frac{\lambda}{\lambda+\varepsilon} \cdot \text{target}_i^{\frac{\varepsilon}{\lambda+\varepsilon}}}{\text{target}_i} = \mu_i^s \frac{\lambda}{\lambda+\varepsilon} \cdot \text{target}_i^{\frac{\varepsilon}{\lambda+\varepsilon}-1} = \mu_i^s \frac{\lambda}{\lambda+\varepsilon} \cdot \text{target}_i^{\frac{-\lambda}{\lambda+\varepsilon}}. \quad (23)$$

The final closed-form update is

$$a_i = \left(\frac{\mu_i^s}{(Kb)_i} \right)^{\frac{\lambda}{\lambda+\varepsilon}}. \quad (24)$$

This formula contains the exponent $\frac{\lambda}{\lambda+\varepsilon}$. When a noisy sample lies far from all experts, all distances are large and K_{ij} becomes very small, which makes the denominator $(Kb)_i$ extremely small. Without this exponent (i.e., in standard Sinkhorn), a_i would blow up to enforce normalization. In As-OT, the exponent < 1 suppresses excessive growth of a_i , so the mass allocated by this sample satisfies $s_i < \mu_i^s$, which implements rejection or discarding. As $\lambda \rightarrow \infty$, the exponent approaches 1 and the method reduces to a hard constraint. When λ is small, the model allows a larger amount of mass loss.

Step 2: Fix a and solve the subproblem over b (expert-side update). We compute the expert-side scaling factor b and enforce strict load balancing across all experts. The subproblem is

$$\max_{b>0} -\mathcal{I}_\infty^*(-\varepsilon \log b) - \varepsilon \langle a, Kb \rangle, \quad (25)$$

which is equivalent to

$$\max_{b>0} \left\{ -\mathcal{I}_\infty^*(-\varepsilon \log b) - \varepsilon \sum_j b_j (K^\top a)_j \right\}. \quad (26)$$

On the expert side, we choose $\mathcal{I}_\infty$ as a strict equality indicator function (hard constraint):

$$\mathcal{I}_\infty(h) = \begin{cases} 0, & h = \nu^s, \\ +\infty, & \text{otherwise.} \end{cases} \quad (27)$$

Its convex conjugate is a linear function. By definition,

$$\mathcal{I}_\infty^*(y) = \sup_t \langle y, h \rangle - \mathcal{I}_\infty(h) = \langle y, \nu^s \rangle. \quad (28)$$

Substituting $y = -\varepsilon \log b$, we obtain

$$\mathcal{I}_\infty^*(-\varepsilon \log b) = \langle -\varepsilon \log b, \nu^s \rangle = -\varepsilon \sum_j \nu_j^s \log b_j. \quad (29)$$

Plugging this back into the subproblem (maximization objective),

$$\max_{b>0} \left\{ -\left(-\varepsilon \sum_j \nu_j^s \log b_j \right) - \varepsilon \sum_j b_j (K^\top a)_j \right\} = \varepsilon \max_{b>0} \left\{ \sum_j \nu_j^s \log b_j - \sum_j b_j (K^\top a)_j \right\}. \quad (30)$$

We drop the constant factor ε and take derivatives with respect to each b_j , then set them to zero:

$$\frac{\partial}{\partial b_j} [\nu_j^s \log b_j - b_j (K^\top a)_j] = \frac{\nu_j^s}{b_j} - (K^\top a)_j = 0. \quad (31)$$

This yields the closed-form update

$$b_j = \frac{\nu_j^s}{(K^\top a)_j}. \quad (32)$$

This is the standard Sinkhorn update. With a simple division, we directly correct the scaling factor b so that the column sums of the final allocation matrix π^s (the total mass received by each expert) strictly match the target distribution ν^s . No matter how the sample side discards mass, the expert side must fully satisfy ν^s . In MoE models, this guarantees that even under strong noise, every expert receives enough data for parameter updates, avoids expert starvation, and achieves expert load balancing.

Overall Sinkhorn Iteration Procedure. Input: cost matrix C , regularization parameter ε , relaxation parameter λ (used on the sample side), marginal distributions μ^s (samples) and ν^s (experts).

1) Initialize $b = \mathbf{1}_M$.

2) While not converged:

- Update sample side (Soft):

$$a \leftarrow \left(\frac{\mu^s}{Kb} \right)^{\frac{\lambda}{\lambda + \varepsilon}} \quad (33)$$

- Update expert side (Hard):

$$b \leftarrow \frac{\nu^s}{K^\top a} \quad (34)$$

3) Compute the final allocation matrix:

$$\pi^s = \text{diag}(a) \cdot K \cdot \text{diag}(b). \quad (35)$$

B.2. The Detailed Derivation of the Dual Formulation

Primal objective:

$$\min_{\pi^s \geq 0} \langle C, \pi^s \rangle + \lambda \text{KL}(\pi^s \mathbf{1}) + \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}) + \varepsilon \sum_{i,j} \pi_{ij}^s (\log \pi_{ij}^s - 1). \quad (36)$$

To derive the dual problem, we first introduce dual potentials for the terms coupled with the row and column marginals. A more direct and commonly used approach is to express the KL term and $\mathcal{I}_\infty$ through their convex conjugates KL^* and $\mathcal{I}_\infty^*$, and then perform element-wise minimization with respect to π^s . Before doing so, we combine the linear term and the entropy term involving π^s into an element-wise separable form, so that we can explicitly compute the inner minimization over π^s .

Combining the linear term with row and column potentials. For dual variables $\phi \in \mathbb{R}^N$ and $\psi \in \mathbb{R}^M$, we have

$$\langle \phi, \pi^s \mathbf{1} \rangle = \sum_{i,j} \phi_i^s \pi_{ij}^s, \quad \langle \psi, \pi^{s\top} \mathbf{1} \rangle = \sum_{i,j} \psi_j \pi_{ij}^s. \quad (37)$$

Therefore,

$$\langle C, \pi^s \rangle + \langle \phi, \pi^s \mathbf{1} \rangle + \langle \psi, \pi^{s\top} \mathbf{1} \rangle = \sum_{i,j} (C_{ij} + \phi_i + \psi_j) \pi_{ij}^s. \quad (38)$$

Including the entropy term, the inner optimization with respect to π^s becomes an element-wise separable problem:

$$\min_{\pi_{ij}^s \geq 0} \sum_{i,j} \left[(C_{ij} + \phi_i + \psi_j) \pi_{ij}^s + \varepsilon \pi_{ij}^s (\log \pi_{ij}^s - 1) \right] + \lambda \text{KL}(\pi^s \mathbf{1}) + \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}) - \langle \phi, \pi^s \mathbf{1} \rangle - \langle \psi, \pi^{s\top} \mathbf{1} \rangle. \quad (39)$$

According to the standard Fenchel duality argument, we can equivalently swap the minimization over π^s and the maximization over ϕ, ψ , which leads to the final dual formulation.

Element-wise minimization of π^s yields a Gibbs-type analytical expression. Fixing ϕ and ψ , we consider the one-dimensional convex problem for each entry π_{ij}^s :

$$\min_{\pi_{ij}^s \geq 0} \pi_{ij}^s (C_{ij} + \phi_i + \psi_j) + \varepsilon \pi_{ij}^s (\log \pi_{ij}^s - 1). \quad (40)$$

This is a convex function of π_{ij}^s . Taking the derivative and setting it to zero yields

$$\frac{d}{d\pi_{ij}^s} [\pi_{ij}^s (C_{ij} + \phi_i + \psi_j) + \varepsilon \pi_{ij}^s (\log \pi_{ij}^s - 1)] = C_{ij} + \phi_i + \psi_j + \varepsilon \log \pi_{ij}^s = 0. \quad (41)$$

Hence,

$$\log \pi_{ij}^s = -\frac{C_{ij} + \phi_i + \psi_j}{\varepsilon}, \quad \pi_{ij}^s = \exp\left(-\frac{C_{ij} + \phi_i + \psi_j}{\varepsilon}\right). \quad (42)$$

In matrix form, for all i, j , we obtain

$$\pi_{ij}^{s*}(\phi, \psi) = \exp\left(-\frac{C_{ij} + \phi_i + \psi_j}{\varepsilon}\right). \quad (43)$$

This is the Gibbs-type solution: each entry is an exponential function of the cost and the dual potentials. The negative exponent maps smaller values of $(C + \phi + \psi)$ to larger allocation weights, and the entropy parameter ε acts as a “temperature” that controls how sharp the distribution is.

Substituting the above equation back, we obtain the Fenchel dual form. Plugging $\pi^{s*}(\phi, \psi)$ back into the inner minimization yields an interaction term

$$-\varepsilon \sum_{i,j} \exp\left(\frac{\phi_i}{\varepsilon}\right) \exp\left(-\frac{C_{ij}}{\varepsilon}\right) \exp\left(\frac{\psi_j}{\varepsilon}\right). \quad (44)$$

We introduce the Gibbs kernel matrix

$$K = \exp\left(-\frac{C}{\varepsilon}\right), \quad K_{ij} = e^{-C_{ij}/\varepsilon}, \quad (45)$$

and use the notation $e^{\phi/\varepsilon}$ for element-wise exponentiation of a vector (and similarly for ψ). The above term can then be written compactly as

$$-\varepsilon \langle e^{\phi/\varepsilon}, K e^{\psi/\varepsilon} \rangle. \quad (46)$$

Expressing KL and $\mathcal{I}_\infty$ through their convex conjugates via the Fenchel transform, the dual problem (maximization over ϕ, ψ) can be written as

$$\max_{a>0, b>0} \left(-\text{KL}^*(-\varepsilon \log a) - \mathcal{I}_\infty^*(-\varepsilon \log b) - \varepsilon \langle a, Kb \rangle \right). \quad (47)$$

This formulation transforms the original matrix optimization problem in $\mathbb{R}^{N \times M}$ into a vector optimization problem over ϕ (length N) and ψ (length M). The dimension reduces from $N \cdot M$ to $N + M$, and the structure becomes an interaction between two vectors coupled through the kernel K , which is well suited for block coordinate ascent.

When solving for π^f , one can simply replace μ^s and ν^s with μ^f and ν^f , respectively.

C. Analysis of the Strict Convexity Introduced by the Entropy Regularization

We analyze the necessity and theoretical implications of introducing the entropic regularization term in the As-OT subproblem of As-COOT.

Convexity of the Unregularized Problem. Consider the As-OT subproblem obtained by fixing π^f and removing the entropic regularization term (the proof for the sub-problem regarding the fixed π^s is similar.):

$$\min_{\pi^s \in \mathbb{R}_+^{N \times M}} \langle C, \pi^s \rangle + \lambda \text{KL}(\pi^s 1_M \mid \mu^s) + \mathcal{I}_\infty\left(\pi^{s\top} 1_N \mid \nu^s\right). \quad (48)$$

The objective in Eq. 48 is convex with respect to π^s , as it consists of a linear cost term, a convex KL divergence over the sample-side marginals, and an indicator function enforcing affine constraints on the expert-side marginals.

However, this objective is *not strictly convex*. In particular, the marginal KL term regularizes only the aggregated distribution $\pi^s 1_M$ rather than the allocation matrix π^s . As a result, multiple allocation schemes sharing identical marginals may attain the same optimal value, yielding a non-singleton set of minimizers.

Consequences of Non-Strict Convexity. The lack of strict convexity in Eq. (48) leads to the following issues: 1) The optimal allocation matrix is not unique. 2) The associated dual problem may be non-smooth and admit non-unique dual variables. 3) When embedded in the nested block coordinate descent (BCD) procedure of As-COOT, the non-uniqueness of subproblem solutions may cause instability or oscillation, undermining convergence guarantees.

Strict Convexity of the Entropic Regularization. We consider the entropic regularization term

$$\text{KL}(\pi^s | \mu^s \otimes \nu^s) = \sum_{i,j} \pi_{ij}^s (\log \pi_{ij}^s - \log(\mu_i^s \nu_j^s)), \quad (49)$$

where $\mu^s \otimes \nu^s$ is fixed.

Define $f(x) = x \log x$ for $x > 0$. Its second derivative satisfies

$$f''(x) = \frac{1}{x} > 0, \quad (50)$$

hence $f(x)$ is strictly convex on $\mathbb{R}_+$.

Since the entropic term can be written as a sum of strictly convex functions $\sum_{i,j} f(\pi_{ij}^s)$ plus a linear term $-\sum_{i,j} \pi_{ij}^s \log(\mu_i^s \nu_j^s)$, the KL divergence $\text{KL}(\pi^s | \mu^s \otimes \nu^s)$ is strictly convex with respect to π^s over $\mathbb{R}_+^{N \times M}$. Therefore, adding the entropic regularization term renders the As-OT subproblem strictly convex and guarantees the uniqueness of the optimal allocation scheme.

Effect of Entropic Regularization. To address these issues, we introduce the entropic regularization term

$$\varepsilon \text{KL}(\pi^s | \mu^s \otimes \nu^s), \quad \varepsilon > 0, \quad (51)$$

which is strictly convex with respect to π^s over $\mathbb{R}_+^{N \times M}$. The resulting regularized objective is given by

$$\min_{\pi^s} \sum_{i,j} \pi_{ij}^s \cdot C_{ij} + \lambda \text{KL}(\pi^s 1_M | \mu^s) + \mathcal{I}_\infty(\pi^{s^\top} 1_N | \nu^s) + \varepsilon \text{KL}(\pi^s | 1). \quad (52)$$

Strict Convexity and Uniqueness. Since the sum of a convex function and a strictly convex function remains strictly convex, the objective in Eq. (52) is strictly convex with respect to π^s . Consequently, the As-OT subproblem admits a unique global minimizer.

Algorithmic Implications. The strict convexity induced by entropic regularization yields several important consequences: 1) The uniqueness of the primal solution stabilizes the nested BCD optimization in As-COOT. 2) the Fenchel–Rockafellar dual of Eq. (52) becomes smooth, leading to a differentiable log-sum-exp form. 3) The optimal allocation matrix admits a Gibbs form and can be efficiently computed via Sinkhorn scaling.

Summary. In As-COOT, the entropic regularization term is not merely introduced for computational convenience. Instead, it plays a fundamental theoretical role by resolving the degeneracy of the transport polytope, ensuring strict convexity and uniqueness of each As-OT subproblem, and providing a principled foundation for stable and globally optimal optimization within the nested asymmetric co-optimal transport framework.

D. Optimality and Convergence of Asymmetric CO-Optimal Transport

1. Problem Setup and Preliminaries

Let $\Omega_s = \{1, \dots, N\}$ and $\Omega_t = \{1, \dots, M\}$ be the index sets for samples and experts, respectively. Let $X = \{x_i\}_{i=1}^N \in \mathbb{R}^{N \times d_1}$ and $E = \{e_i\}_{i=1}^M \in \mathbb{R}^{M \times d_2}$ represent the feature matrices. The cost tensor is denoted by $C = L(X, E) \in \mathbb{R}^{N \times M \times d_1 \times d_2}$.

We define the set of positive measures on discrete spaces as $\mathcal{M}_+(\Omega)$. The Asymmetric CO-Optimal Transport problem seeks a pair of couplings (π^s, π^f) minimizing a joint cost with asymmetric marginal constraints: soft constraints on the sample side and hard constraints on the expert side.

Definition 1 (As-COOT Objective Function). The objective function $\mathcal{J} : \mathbb{R}_+^{N \times M} \times \mathbb{R}_+^{d_1 \times d_2} \rightarrow \mathbb{R} \cup \{+\infty\}$ is defined as:

$$\mathcal{J}(\pi^s, \pi^f) = \langle C, \pi^s \otimes \pi^f \rangle + \mathcal{R}_s(\pi^s) + \mathcal{R}_f(\pi^f) + \varepsilon H(\pi^s \otimes \pi^f | \mu \otimes \nu) \quad (53)$$

where:

- The coupling term is $\langle C, \pi^s \otimes \pi^f \rangle = \sum_{i,j,k,l} L_{ijkl} \pi_{ij}^s \pi_{kl}^f$.
- $\mathcal{R}_s(\pi^s) = \lambda \text{KL}(\pi^s \mathbf{1}_M | \mu^s) + \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}_N | \nu^s)$ represents the asymmetric constraint (Soft Source / Hard Target).
- $\mathcal{R}_f(\pi^f)$ is the analogous regularization for the feature alignment matrix.
- $H(\pi | \xi) = \sum \pi \log(\pi / \xi) - \pi + \xi$ is the generalized Kullback-Leibler divergence used for entropic regularization.

2. Existence of Global Minimizers

Theorem 1 (Existence). The As-COOT optimization problem admits at least one global minimizer (π_*^s, π_*^f) .

Proof. Let the feasible set be $\mathcal{F} = \{(\pi^s, \pi^f) \in \mathbb{R}_+^{N \times M} \times \mathbb{R}_+^{d_1 \times d_2} \mid \pi^{s\top} \mathbf{1}_N = \nu^s, \pi^{f\top} \mathbf{1}_{d_1} = \nu^f\}$.

Compactness of Feasible Set: the constraint $\pi^{s\top} \mathbf{1}_N = \nu^s$ implies that for each column j , $\sum_i \pi_{ij}^s = \nu_j^s$. Since $\pi_{ij}^s \geq 0$, each element is bounded by $0 \leq \pi_{ij}^s \leq \nu_j^s$. Thus, π^s belongs to a closed and bounded subset of Euclidean space, which is compact by the Heine-Borel theorem. The same applies to π^f . Therefore, the product space $\mathcal{F}$ is compact.

Lower Semi-Continuity (L.S.C.):

- The bilinear term $\langle C, \pi^s \otimes \pi^f \rangle$ is continuous w.r.t. (π^s, π^f) .
- The KL divergence is continuous on its domain and strictly convex.
- The indicator function $\mathcal{I}_\infty$ is L.S.C. (it is the convex indicator of a closed set).
- The sum of L.S.C. functions is L.S.C.

By the Weierstrass Extreme Value Theorem, a lower semi-continuous function on a non-empty compact set attains its minimum.

3. Sub-problem Analysis: Convexity and Duality

Since the objective is non-convex jointly but bi-convex, we analyze the sub-problem where one variable is fixed. Without loss of generality, fix π^f and let $C_{ij}^s = \sum_{k,l} C_{ijkl} \pi_{kl}^f$.

Problem $\mathcal{P}_s(\pi^f)$:

$$\min_{\pi^s \geq 0} \langle C^s, \pi^s \rangle + \lambda \text{KL}(\pi^s \mathbf{1}_M | \mu^s) + \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}_N) + \varepsilon \text{KL}(\pi^s | \mu^s \otimes \nu^s) \quad (54)$$

1) Strict Convexity. Problem $\mathcal{P}_s(\pi^f)$ is strictly convex w.r.t π^s .

Proof. The term $\varepsilon \text{KL}(\pi^s | \mu^s \otimes \nu^s)$ involves the function $x \log x$, which has a positive second derivative ($1/x > 0$ for $x > 0$). The linear terms and the indicator function are convex. The sum of a strictly convex function and convex functions is strictly convex. Thus, if a solution exists, it is unique.

2) Fenchel-Rockafellar Dual Derivation. To derive the optimal update law, we construct the Lagrangian dual. Let $g(\pi^s) = \lambda \text{KL}(\pi^s \mathbf{1}_M | \mu^s)$ and $r(\pi^s) = \mathcal{I}_\infty(\pi^{s\top} \mathbf{1}_N)$. The primary problem is:

$$\min_{\pi^s} \mathcal{E}(\pi^s) + g(A\pi^s) + r(B\pi^s) \quad (55)$$

where $\mathcal{E}$ is the entropic cost, A sums rows, and B sums columns. Using Fenchel duality, the dual problem is:

$$\max_{\alpha \in \mathbb{R}^N, \beta \in \mathbb{R}^M} -g^*(-\alpha) - r^*(-\beta) - \varepsilon \sum_{i,j} \exp\left(\frac{\alpha_i + \beta_j - C_{ij}^s}{\varepsilon}\right) \quad (56)$$

where g^* and h^* are the Legendre-Fenchel conjugates:

- Soft Constraint Conjugate (g^*): For $g(x) = \lambda \sum (x_i \log(x_i/\mu_i^s) - x_i + \mu_i^s)$, the conjugate is:

$$g^*(y) = \lambda \sum_i \mu_i^s \left(e^{y_i/\lambda} - 1 \right) \quad (57)$$

- Hard Constraint Conjugate (r^*): For $r(x) = \mathcal{I}_\infty(x)$, the conjugate is linear:

$$r^*(y) = \langle y, \nu^s \rangle \quad (58)$$

Substituting these into the dual objective:

$$\max_{\alpha, \beta} -\lambda \sum_i \mu_i^s \left(e^{-\alpha_i/\lambda} - 1 \right) + \langle \beta, \nu^s \rangle - \varepsilon \sum_{i,j} \mu_i^s \nu_j^s e^{(\alpha_i + \beta_j - C_{ij}^s)/\varepsilon} \quad (59)$$

3) Optimality Conditions (KKT). Setting the gradient of the dual objective w.r.t α_i and β_j to zero yields the optimality conditions.

For α_i :

$$\mu_i^s e^{-\alpha_i/\lambda} = \sum_j \mu_i^s \nu_j^s e^{(\alpha_i + \beta_j - C_{ij}^s)/\varepsilon} \quad (60)$$

$$e^{-\alpha_i/\lambda} = e^{\alpha_i/\varepsilon} \sum_j K_{ij} \nu_j^s e^{\beta_j/\varepsilon} \quad (61)$$

where $K_{ij} = e^{-C_{ij}^s/\varepsilon}$

Let $u_i = e^{\alpha_i/\varepsilon}$ and $v_j = e^{\beta_j/\varepsilon}$. The equation becomes:

$$u_i^{-\varepsilon/\lambda} = u_i \sum_j K_{ij} \nu_j^s v_j \quad (62)$$

Rearranging for u_i :

$$u_i = \left(\sum_j K_{ij} \nu_j^s v_j \right)^{-\frac{\lambda}{\lambda + \varepsilon}} \quad (63)$$

For β_j (Hard constraint):

$$\nu_j^s = \sum_i \mu_i^s \nu_j^s e^{(\alpha_i + \beta_j - C_{ij}^s)/\varepsilon} \quad (64)$$

$$1 = v_j \sum_i u_i K_{ij} \mu_i^s \implies v_j = \frac{1}{\sum_i \mu_i^s K_{ij} u_i} \quad (65)$$

Theorem 2 (Optimal Sub-problem Solution). The unique global minimizer π_*^s for the fixed- π^f sub-problem is given by:

$$\pi_{ij}^s = u_i K_{ij} v_j \mu_i^s \nu_j^s \quad (66)$$

where vectors u and v are determined by the fixed-point iterations of the generalized Sinkhorn algorithm derived above.

4. Convergence of Block Coordinate Descent (BCD)

The As-COOT algorithm alternates between solving for π^s and π^f . We model this as a Block Coordinate Descent on the objective $\mathcal{J}$.

Algorithm Flow:

$$\pi_{k+1}^s = \arg \min_{\pi^s} \mathcal{J}(\pi^s, \pi_k^f) \quad (67)$$

$$\pi_{k+1}^f = \arg \min_{\pi^f} \mathcal{J}(\pi_{k+1}^s, \pi^f) \quad (68)$$

Theorem 3 (Monotonicity and Convergence). The sequence $\{(\pi_k^s, \pi_k^f)\}_{k \in \mathbb{N}}$ generated by As-COOT satisfies. Monotonicity: $\mathcal{J}(\pi_{k+1}^s, \pi_{k+1}^f) \leq \mathcal{J}(\pi_k^s, \pi_k^f)$. Stationarity: Every limit point of the sequence is a stationary point of the problem.

Proof. Descent Property: by definition of the update steps:

$$\mathcal{J}(\pi_{k+1}^s, \pi_k^f) \leq \mathcal{J}(\pi_k^s, \pi_k^f) \quad (69)$$

since π_{k+1}^s is the global minimizer

$$\mathcal{J}(\pi_{k+1}^s, \pi_{k+1}^f) \leq \mathcal{J}(\pi_{k+1}^s, \pi_k^f) \quad (70)$$

since π_{k+1}^f is the global minimizer

Summing these inequalities confirms $\mathcal{J}_{k+1} \leq \mathcal{J}_k$. Since $\mathcal{J}$ is bounded below (by 0), the sequence of objective values converges.

Convergence to Stationary Point: the objective function $\mathcal{J}$ can be decomposed as:

$$\mathcal{J}(\pi^s, \pi^f) = f(\pi^s, \pi^f) + \sum_i g_i(\pi^s) + \sum_j r_j(\pi^f) \quad (71)$$

where f is the smooth differentiable part (coupling term + entropic parts treated smoothly in the interior) and g, r are the non-smooth convex indicator functions for constraints.

According to Tseng's Theorem (Tseng, 2001) on Coordinate Descent for non-smooth separable functions: 1) The non-smooth parts are separable (indicator on π^s does not depend on π^f). 2) The smooth part is regular on the domain. 3) The sub-problems maximize a strictly concave dual (or minimize a strictly convex primal) and have unique solutions. Under these conditions, any limit point $(\bar{\pi}^s, \bar{\pi}^f)$ of the sequence satisfies the first-order optimality conditions (stationary point):

$$0 \in \partial_{\pi^s} \mathcal{J}(\bar{\pi}^s, \bar{\pi}^f) \quad \text{and} \quad 0 \in \partial_{\pi^f} \mathcal{J}(\bar{\pi}^s, \bar{\pi}^f) \quad (72)$$

Thus, As-COOT converges to a coordinate-wise minimum.

5. Summary

We have rigorously established the following properties for As-COOT. 1) Well-Posedness: The problem is defined on a compact set with a lower semi-continuous objective, guaranteeing the existence of an optimal solution (*Theorem 1*). 2) Tractability: The non-convex joint problem decomposes into strictly convex sub-problems. Through Fenchel duality, we proved that the sub-problems admit a unique solution expressible via efficient generalized Sinkhorn scalings (*Theorem 2*). 3) Algorithmic Convergence: The Block Coordinate Descent strategy is theoretically guaranteed to monotonically decrease the objective and converge to a stationary point (*Theorem 3*).

This theoretical foundation confirms that As-COOT is not merely a heuristic, but a mathematically sound optimization framework for asymmetric distribution alignment.

E. Reduction from As-COOT to As-OT under Block Coordinate Descent

Recall that the As-COOT objective is defined as

$$\min_{\pi_t^s, \pi_t^f} \langle L(X_t, E_t), \pi_t^s, \pi_t^f \rangle + \lambda KL(\pi_t^s 1_M \otimes \pi_t^f 1_{d_2} \mid \mu) + \mathcal{I}_\infty(\pi_t^s{}^\top 1_N \otimes \pi_t^f{}^\top 1_{d_1} \mid \nu) + \varepsilon KL(\pi_t^s \otimes \pi_t^f \mid \mu \otimes \nu). \quad (73)$$

Since the objective in Eq. 73 is jointly non-convex with respect to (π_t^s, π_t^f) , we adopt a block coordinate descent (BCD) strategy to solve it. Specifically, when fixing π_t^f and optimizing with respect to π_t^s , the As-COOT objective reduces to the following subproblem:

$$\min_{\pi_t^s} \sum_{i,j} \underbrace{\pi_{t,ij}^s \sum_{k,l} \pi_{t,kl}^f L(X_t(i,k), E_t(j,l))}_{\triangleq C_{ij}} + \lambda KL(\pi_t^s 1_M \mid \mu^s) + \mathcal{I}_\infty(\pi_t^s{}^\top 1_N \mid \nu^s) + \varepsilon KL(\pi_t^s \mid \mu^s \otimes \nu^s), \quad (74)$$

Here, the bilinear cost term in As-COOT collapses into a linear allocation cost $\langle C, \pi_t^s \rangle$, where the effective cost matrix $C \in \mathbb{R}^{N \times M}$ is defined as

$$C_{ij} = \sum_{k,l} \pi_{t,kl}^f L(X_t(i, k), E_t(j, l)). \quad (75)$$

Moreover, since π_t^f is fixed, the marginal relaxation on the sample side and the hard marginal constraint on the expert side depend only on π_t^s . Therefore, Eq. (74) is exactly an *asymmetric optimal transport (As-OT)* problem with a soft marginal constraint on the sample side and a hard marginal constraint on the expert side.

Simplification of the Entropic Regularization. Recall that the entropic regularization term in As-COOT is defined as

$$\varepsilon \text{KL}(\pi_t^s \otimes \pi_t^f \mid \mu \otimes \nu) = \varepsilon \sum_{(i,k),(j,l)} (\pi_t^s \otimes \pi_t^f)_{(i,k),(j,l)} \log \frac{(\pi_t^s \otimes \pi_t^f)_{(i,k),(j,l)}}{(\mu \otimes \nu)_{(i,k),(j,l)}}. \quad (76)$$

Under BCD, when fixing π_t^f and optimizing with respect to π_t^s , the joint coupling $\pi_t^s \otimes \pi_t^f$ factorizes, and the KL divergence in Eq. (76) can be expanded as

$$\begin{aligned} \text{KL}(\pi_t^s \otimes \pi_t^f \mid \mu \otimes \nu) &= \sum_{i,j,k,l} \pi_{t,ij}^s \pi_{t,kl}^f \log \frac{\pi_{t,ij}^s \pi_{t,kl}^f}{\mu_i^s \mu_k^f \nu_j^s \nu_l^f} \\ &= \sum_{i,j,k,l} \pi_{t,ij}^s \pi_{t,kl}^f (\log \pi_{t,ij}^s - \log \mu_i^s - \log \nu_j^s) + \sum_{i,j,k,l} \pi_{t,ij}^s \pi_{t,kl}^f (\log \pi_{t,kl}^f - \log \mu_k^f - \log \nu_l^f). \end{aligned} \quad (77)$$

Since π_t^f is fixed in the current subproblem, the second term can be rewritten as

$$\sum_{i,j} \pi_{t,ij}^s \sum_{k,l} \pi_{t,kl}^f (\log \pi_{t,kl}^f - \log \mu_k^f - \log \nu_l^f),$$

which is equal to a constant due to the constraint $\sum_{i,j} \pi_{t,ij}^s = 1$, and therefore reduces to an additive constant, which can be safely discarded without affecting the optimizer.

The remaining term yields

$$\varepsilon \sum_{i,j} \pi_{t,ij}^s (\log \pi_{t,ij}^s - \log \mu_i^s - \log \nu_j^s) = \varepsilon \text{KL}(\pi_t^s \mid \mu^s \otimes \nu^s). \quad (78)$$

Moreover, the linear terms $-\varepsilon \sum_{i,j} \pi_{t,ij}^s (\log \mu_i^s + \log \nu_j^s)$ can be absorbed into the cost matrix or, equivalently, into the dual potentials. Therefore, up to additive constants and linear reparameterizations that do not affect the optimal coupling, the entropic regularization term $\varepsilon \text{KL}(\pi_t^s \mid \mu^s \otimes \nu^s)$ is equivalent to the negative entropy regularization

$$\varepsilon \text{KL}(\pi_t^s \mid \mathbf{1}), \quad (79)$$

which yields the standard Sinkhorn scaling form.

Rewriting the joint entropic regularization term $\varepsilon \text{KL}(\pi_t^s \otimes \pi_t^f \mid \mu \otimes \nu)$ into the reduced form $\varepsilon \text{KL}(\pi_t^s \mid \mathbf{1})$ under block coordinate descent brings several practical and theoretical advantages. First, it preserves the optimal solution of the original subproblem up to additive constants, since all discarded terms are independent of the optimized variable or can be absorbed into the cost or dual potentials. Second, the reduced formulation yields a strictly convex objective with respect to π_t^s , guaranteeing the existence and uniqueness of the subproblem solution. Finally, this form incorporates the standard Sinkhorn scaling updates with a Gibbs kernel, allowing for efficient and numerically stable optimization with linear-time iterations, which would be challenging to achieve when keeping the full reference measure explicitly.

The Final As-OT Form:

$$\min_{\pi_t^s} \sum_{i,j} \pi_{t,ij}^s \cdot C_{ij} + \lambda \text{KL}(\pi_t^s \mathbf{1}_M \mid \mu^s) + \mathcal{I}_\infty(\pi_t^s \mathbf{1}_N \mid \nu^s) + \varepsilon \text{KL}(\pi_t^s \mid \mathbf{1}). \quad (80)$$

F. Analysis of Noisy Samples

F.1. Noisy Samples in CTTA: Definition and Rationale

In CTTA, models are exposed to an unlabeled and continuously evolving test stream, where domain shifts accumulate over time. Unlike standard i.i.d. test-time inference, CTTA inevitably encounters a substantial portion of samples that are unreliable for adaptation. Prior CTTA studies have observed this phenomenon under different terms, such as low-confidence samples (Wang et al., 2023), hard samples (Zuo et al., 2021), or unreliable pseudo-labeled (Wang et al., 2022) instances, and have shown that adapting on all test samples often leads to error accumulation and performance degradation.

In this work, we define noise in CTTA explicitly as samples with high allocation cost. From a functional and allocation-centric perspective, a test sample is regarded as noisy if it cannot be reliably assigned to any expert under the current model state. Such samples typically arise during domain transitions, at distribution boundaries, or in regions where the evolving test distribution is insufficiently covered by the existing expert functionalities. Importantly, they are not necessarily out-of-distribution or permanently uninformative; rather, they are temporarily incompatible with the current expert ensemble. After SSA, the sample–expert allocation cost directly reflects this functional incompatibility. Consequently, a sample that incurs consistently high allocation costs across all experts is difficult to explain by the current expert set and cannot be confidently routed for adaptation. This cost-based definition of noise is consistent with prior findings in optimal transport–based learning: Feng et al. (2023) shows that samples with persistently high allocation costs are strong indicators of noisy supervision, and that forcing their alignment degrades overall learning dynamics. If high-cost samples are forced to be allocated, it has two adverse effects in CTTA: (1) due to the global coupling nature of allocation optimization, they distort the allocation of clean samples, leading to routing errors, and (2) they contaminate expert updates during online model adaptation.

As-COOT addresses this issue by relaxing the sample-side marginal constraint via a KL divergence penalty, allowing allocation mass to be partially discarded at a controlled cost. This mechanism enables the model to downweight or temporarily ignore high-cost samples. We emphasize that this discarding mechanism does not reduce inference coverage or permanently exclude samples; instead, it prevents unreliable samples from dominating routing and adaptation at stages when they are not yet compatible with the expert ensemble. This design choice is validated by our ablation studies (as § 4.2). Replacing As-COOT with standard COOT, removing the noise-aware discarding mechanism, leads to consistent performance degradation, demonstrating that explicitly accounting for noisy, high-cost samples is critical for stable and effective CTTA. Together, these analyses justify our treatment of high allocation cost samples as noise and highlight the necessity of As-COOT in CTTA.

F.2. Noise Robustness Analysis of As-COOT Compared to COOT

Consider a noisy sample i , whose cost with respect to all experts is high:

$$C_{ij} \gg 0, \forall j. \quad (81)$$

The corresponding Gibbs kernel entries satisfy

$$K_{ij} = \exp\left(-\frac{C_{ij}}{\varepsilon}\right) \approx 0. \quad (82)$$

Case of COOT. In COOT, a hard marginal constraint is imposed on the sample side:

$$s_i = \sum_j \pi_{ij}^s = \frac{1}{N}, \quad (83)$$

which must be satisfied regardless of the magnitude of C_{ij} .

When updating the sample-side scaling factor, we have

$$a_i = \frac{\mu_i^s}{(Kb)_i}, \quad (Kb)_i = \sum_j K_{ij} b_j. \quad (84)$$

For a noisy sample, $K_{ij} \approx 0$ for all j , leading to $(Kb)_i \approx 0$ and thus

$$a_i \rightarrow \infty. \quad (85)$$

As a consequence, the allocation scheme

$$\pi_{ij}^s = a_i K_{ij} b_j \quad (86)$$

is forced to adopt an extremely large a_i in order to satisfy the hard constraint $s_i = \frac{1}{N}$. This implies that noisy samples are still compulsorily allocated to experts. To accommodate this constraint, the allocations of clean samples are distorted, resulting in an increased overall allocation cost induced by noise.

Case of As-COOT. In contrast, As-COOT adopts a KL-divergence-based soft marginal constraint on the sample side. The corresponding update rule is given by

$$a_i = \left(\frac{\mu_i^s}{(Kb)_i} \right)^{\frac{\lambda}{\lambda+\epsilon}}. \quad (87)$$

When $(Kb)_i \approx 0$ (i.e., for a noisy sample):

- For finite λ , the exponent $\frac{\lambda}{\lambda+\epsilon} < 1$ suppresses the amplification effect of the scaling factor;
- The resulting total allocated mass is

$$s_i = a_i \cdot (Kb)_i = \mu_i^s \frac{\lambda}{\lambda+\epsilon} \cdot (Kb)_i^{\frac{\epsilon}{\lambda+\epsilon}}. \quad (88)$$

Since $(Kb)_i \approx 0$ and $\frac{\epsilon}{\lambda+\epsilon} > 0$, it follows that

$$s_i \rightarrow 0. \quad (89)$$

This indicates that the total mass allocated to noisy samples in As-COOT naturally vanishes, effectively discarding them from the allocation scheme. Therefore, As-COOT achieves noise robustness by softly relaxing the marginal constraint, implicitly performing adaptive sample suppression for high-cost noisy inputs.

G. Computational Complexity Analysis of As-COOT

We analyze the computational complexity of solving As-COOT, with particular emphasis on its *effective runtime* under GPU-parallel execution.

Algorithmic Structure. As-COOT solves the joint optimization over (π_t^s, π_t^f) via a block coordinate descent (BCD) scheme. At each outer iteration, the algorithm alternates between two Asymmetric Optimal Transport (As-OT) subproblems: (i) fixing π_t^f and updating π_t^s , and (ii) fixing π_t^s and updating π_t^f . Each As-OT subproblem is solved using Sinkhorn scaling with entropic regularization.

Let T_{out} denote the number of outer BCD iterations and T_{in} the number of Sinkhorn iterations.

Cost Matrix Construction. At each outer iteration, As-COOT constructs two cost matrices: $C_s \in \mathbb{R}^{N \times M}$ and $C_f \in \mathbb{R}^{d_1 \times d_2}$, where N and M are the numbers of samples and experts, and d_1, d_2 are the feature dimensions of X_t and E_t , respectively. The dominant operations in computing C_s and C_f are dense matrix multiplications, such as $X_t \pi_t^f E_t^\top$ and $X_t^\top \pi_t^s E_t$.

While the arithmetic complexity of these operations is cubic in the dimensions, they are implemented using GPU-optimized GEMM kernels. Under a model with sufficient computational resources, their *parallel time complexity* is

$$\mathcal{O}(\max\{N, M, d_1, d_2\}). \quad (90)$$

Asymmetric Sinkhorn Iterations. Each As-OT subproblem is solved using Sinkhorn scaling. At each Sinkhorn iteration, the updates

$$a \leftarrow \left(\mu^{s/f} \odot (Kb) \right)^\gamma, \quad b \leftarrow \nu^{s/f} \odot (K^\top a) \quad (91)$$

involve matrix–vector multiplications, element-wise divisions, and exponentiation. These operations correspond to parallel reductions and element-wise kernels on GPUs, whose parallel depth is constant.

As a result, each Sinkhorn iteration has $\mathcal{O}(1)$ parallel time complexity, and solving one As-OT subproblem costs

$$\mathcal{O}(T_{\text{in}}). \quad (92)$$

Overall Parallel Time Complexity. Combining the cost matrix construction and the two As-OT updates, the overall *parallel time complexity* of As-COOT is

$$\mathcal{O}\left(T_{\text{out}} \cdot [T_{\text{in}} + \max\{N, M, d_1, d_2\}]\right). \quad (93)$$

In practice, both T_{out} and T_{in} are small constants (e.g., $T_{\text{out}} \leq 20$, $T_{\text{in}} \leq 100$). Therefore, the effective runtime of As-COOT scales linearly with the largest problem dimension.

As-COOT avoids combinatorial optimization, sorting, or discrete allocation steps. All computations consist of dense matrix multiplications and element-wise operations, which are highly optimized and fully parallelizable on GPUs.

H. Experimental Setting Details

H.1. Implementation Details

We conduct experiments on Qwen1.5-MoE-A2.7B-Chat (Team, 2024) and DeepSeek-V2-Lite (Liu et al., 2024a). Qwen1.5-MoE-A2.7B-Chat is a Mixture-of-Experts (MoE) model with 14.3B total parameters and 2.7B activated parameters at runtime; it uses 24 MoE layers, each with 60 experts and a shared expert per MoE layer. DeepSeek-V2-Lite is a 15.7B-parameter MoE model with 2.4B activated parameters, comprising 1 MLP layer and 26 MoE layers, where each MoE layer is composed of 64 routed experts and 2 shared experts. We use all-MiniLM-L6-v2 as the sample encoder and a two-layer MLP with GELU activation and ℓ_2 normalization as the expert encoder, where the expert encoder takes as input the flattened LoRA adapter parameters extracted from each expert at a given layer (encoders are mentioned in Sec. 3.2). Both encoders are trained with temperature $\tau = 0.07$. We optimize the sample encoder and the expert encoder jointly using the Adam optimizer (Kingma & Ba, 2017). LoRA adapters with rank $r = 16$ and scaling factor $\alpha = 16$ are injected into all expert MLP modules, and their outputs are added to the original expert activations. To ensure fair comparison and reduce computational overhead, all methods uniformly inject LoRA adapters when using MoE-based models. All experiments are performed on a single NVIDIA A800 GPU. We use the AdamW optimizer (Loshchilov & Hutter, 2017) with a batch size of 8 for Qwen and 4 for DeepSeek, and a learning rate of $1e-5$. For expert routing, we activate the top-2 experts per MoE layer. Unless otherwise specified, the hyperparameters are set to $\alpha = 5.0$, $\lambda = 0.1$, and $p = 0.01$ across all experiments. By default, our TiME is enabled at the first MoE layer of the model (layer 0 for Qwen and layer 1 for DeepSeek), and is implemented using only a single MoE layer. The performance and analysis of TiME across different MoE layers are provided in the App. O.5.

H.2. Detailed Explanation of As-COOT-based Test-Time Routing Calibration

We perform routing calibration in the logit space as $z_{\text{As}} = z_o + \alpha \cdot \log(\widetilde{\pi}_t^s + \epsilon)$, where $\widetilde{\pi}_t^s$ provides a domain-level prior over expert allocation for the current sample. This calibration is motivated by the mismatch between the granularity at which domain information is estimated and the granularity at which MoE routing is performed. Specifically, As-COOT produces a sample-level (i.e., domain-level) allocation, whereas the MoE router operates at the token level based on token representations. Consequently, an explicit mechanism is required to incorporate domain-level information into token-level routing decisions.

Performing calibration in the logit space enables the domain-level prior to be incorporated in a principled and numerically stable manner. By adding the logarithm of $\widetilde{\pi}_t^s$ to the original routing logits, the calibrated routing distribution is equivalent, after softmax, to multiplicatively reweighting the original token-level routing probabilities by a domain-level routing prior. This formulation can be interpreted as introducing a Bayesian-style prior over experts conditioned on the current domain, while preserving the relative preferences induced by token-level features. Since each row of $\widetilde{\pi}_t^s$ corresponds to a domain-level routing distribution for a given sample, whereas MoE routing is performed independently for each token, we broadcast the corresponding row of $\widetilde{\pi}_t^s$ across all tokens within the same sample and add it to their routing logits. In this way, all tokens share a consistent domain-aware bias over expert allocation, while still allowing fine-grained token-level variations to be captured by the original router. The calibration strength α controls the influence of the domain-level prior, and ϵ ensures numerical stability by preventing undefined logarithms.

H.3. Details of the Benchmark Datasets

Based on prior work (Liu et al., 2025; Chen et al., 2025a), we construct the CTTA benchmark using the EleutherAI Language Model Evaluation Harness (Gao et al., 2024). The benchmark includes two task streams: text understanding (TU) and

medical reasoning (**Med**).

We illustrate the details of each dataset in the **Text Understanding (TU)** task stream as follows.

- **AI2 Reasoning Challenge (ARC)** (Clark et al., 2018) consists of 7,787 science exam questions drawn from a variety of sources, including questions provided under license by a research partner affiliated with AI2. Each question follows a multiple-choice format with four answer options. The dataset is divided into two subsets. The Challenge (ARC-c) Set contains 2,590 hard questions, defined as those that cannot be correctly answered by either a retrieval-based method or a co-occurrence-based method. The Easy Set (ARC-e) includes the remaining 5,197 questions.
- **BoolQ** (Clark et al., 2019) is a question answering task in which each example consists of a short passage paired with a binary (yes/no) question. The questions are anonymously and organically collected from users of the Google Search engine and are subsequently matched with relevant paragraphs from Wikipedia articles that contain the answers.
- **OpenBookQA (OBQA)** (Mihaylov et al., 2018) is a question answering dataset designed to assess subject understanding in an open-book exam setting. It consists of multiple-choice elementary-level science questions that evaluate a model’s ability to apply a small set of core science facts to novel situations. Each question is associated with a corresponding core fact that it is intended to probe.
- **Recognizing Textual Entailment (RTE)** (Bentivogli et al., 2009) consists of pairs of short text fragments, each composed of a text and a corresponding hypothesis. Each pair is annotated with a binary label indicating whether the hypothesis can be inferred from the given text, capturing fine-grained semantic relationships between sentence pairs.
- **Cross-lingual Question Answering Dataset (XQuAD)** (Dumitrescu et al., 2021) is a cross-lingual QA dataset derived from the development set of SQuAD v1.1 (Rajpurkar et al., 2016), consisting of question–answer pairs aligned with their corresponding context paragraphs. The dataset provides translated versions across multiple languages. Following prior work (Liu et al., 2025), we select the Chinese (XQuAD-Zh) and Spanish (XQuAD-Es) subsets.

We illustrate the details of each dataset in the **Medical Reasoning (Med)** task stream as follows.

- **MedConceptsQA** (Shoham & Rappoport, 2024) is a medical question answering dataset focused on medical concepts across multiple standardized vocabularies. It consists of questions related to core medical knowledge, covering diagnoses, procedures, and drugs. It includes five subsets: ICD9-CM focuses on diagnosis codes defined in the ICD-9 Clinical Modification, containing questions related to disease classification and clinical condition descriptions under the ICD-9 standard. ICD10-CM covers diagnosis codes from the ICD-10 Clinical Modification, which provides a more fine-grained and expanded coding system, showing more detailed clinical distinctions. ICD9-PROC consists of questions associated with medical procedure codes defined in the ICD-9 Procedure Coding System, emphasizing procedural classifications used in earlier clinical coding practices. ICD10-PROC contains questions related to procedure codes from the ICD-10 Procedure Coding System, which introduces a more expressive and structured representation of medical procedures. The ATC focuses on the Anatomical Therapeutic Chemical classification system, including drug categories organized by therapeutic use, pharmacological properties, and chemical characteristics.
- **MedQA** (Jin et al., 2021) is an open-domain medical question answering dataset composed of questions drawn from medical board examinations. The questions originate from professional medical exams and are designed to assess medical knowledge and clinical decision-making ability.
- **MedMCQA** (Pal et al., 2022) is a multiple-choice medical question answering dataset composed of questions collected from AIIMS and NEET PG medical entrance examinations. The dataset covers a wide range of medical subjects and healthcare topics. Each instance includes a question, the correct answer, and multiple distractor options.
- **PubMedQA** (Jin et al., 2019) is a biomedical question answering dataset constructed from PubMed abstracts. Each instance consists of a research-style question paired with an abstract context and is annotated with a binary or ternary answer (yes/no/maybe) that summarizes the abstract’s conclusion. The dataset includes expert-annotated, unlabeled, and automatically generated question–answer instances.

H.4. Evaluation Metrics

All evaluation metrics of each dataset are selected following the default settings of the EleutherAI Language Model Evaluation Harness (Gao et al., 2024).

For Winogrande, ARC-c, ARC-e, BoolQ, OpenBookQA (OBQA), RTE in the Text Understanding (TU) task stream, and MedConceptsQA, MedQA, MedMCQA, PubMedQA in the Medical Reasoning (Med) task stream, we report **Accuracy (Acc)**, defined as the proportion of correctly predicted labels over all evaluation instances.

For XQuAD-Zh and XQuAD-Es in the TU, we adopt **Exact Match (EM)** (Rajpurkar et al., 2016). EM measures the percentage of predictions that exactly match the ground-truth answer after standard normalization (e.g., lowercasing and punctuation removal).

H.5. Baselines

We compare our TiME with mainstream CTTA methods. *Tent* (Wang et al., 2021) updates the BN parameters via minimizing the prediction entropy. *CoTTA* (Wang et al., 2022) combines a teacher-student framework and data augmentation. *SAR* (Niu et al., 2023) uses sharpness-aware minimization to find a flatter loss space. *SoTTA* (Gong et al., 2023) utilizes a sharpness-aware optimizer to improve the reliability of model updates and facilitate adaptation to highly challenging environments. *DSS* (Wang et al., 2024b) uses a teacher-student framework and mitigates noise through positive-negative learning. *REM* (Han et al., 2025) enforces difficulty-aware adaptation via ranking masked consistency. *CTTA-T* (Liu et al., 2025) uses a teacher-student framework to perceive domain shift and refine-then-filter noisy sample via prediction consistency.

We compare our TiME with mainstream online MoE re-routing methods. *THOR* (Zuo et al., 2022) randomly allocates experts to each input with consistency. *Rewiring* (Su et al., 2025) leverages self-supervision via lightweight router logit updates to rewire expert selection during testing. *R2-T2* (Li et al., 2025) leverages optimization guided by reference samples, performing expert re-routing without additional training during testing.

Table 4. Comparison of different CTTA methods on the Medical Reasoning (Med) task stream using Qwen1.5-MoE-A2.7B-Chat. **TiME (EOE)** uses expert output entropy, while **TiME (OR)** adopts the original routing. **Bold** and Underline denote the best and second-best results, respectively. Source denotes zero-shot performance without adaptation.

Base Model	Method	$t \rightarrow$								Avg $\uparrow$
		ICD9-CM Acc	ICD10-CM Acc	ATC Acc	ICD9-PROC Acc	ICD10-PROC Acc	MedQA Acc	MedMCQA Acc	PubMedQA Acc	
Qwen1.5-MoE-A2.7B	Source	0.2638	0.2407	0.2373	0.2609	0.2571	0.3371	0.3173	0.9464	0.3576
	Tent	0.2538	0.2390	0.2542	0.2483	0.2564	0.3055	0.2881	0.9107	0.3445
	CoTTA	0.2411	0.2335	0.2539	0.2436	0.2652	0.3020	0.3045	0.9201	0.3455
	SAR	0.2741	0.2440	0.2542	0.2554	0.2516	0.3109	0.2851	0.9423	0.3522
	SoTTA	0.2386	0.2677	0.2573	0.2648	0.2394	0.3446	0.3329	0.9285	0.3592
	DSS	0.2638	0.2512	0.2347	0.2563	0.2525	0.3477	0.3236	0.9401	0.3588
	REM	0.2525	0.2601	0.2512	0.2525	0.2449	0.3020	0.2500	0.9032	0.3396
	CTTA-T	<u>0.2804</u>	0.2563	0.2373	0.2576	0.2563	0.3493	0.3083	0.9401	0.3607
	TiME (EOE)	0.2664	0.2591	0.2643	<u>0.2637</u>	<u>0.2715</u>	0.3518	0.3783	<u>0.9481</u>	<u>0.3754</u>
	TiME (OR)	0.2826	<u>0.2639</u>	<u>0.2579</u>	0.2618	0.2803	<u>0.3506</u>	<u>0.3594</u>	0.9505	0.3759

I. Evaluation of Time and Memory Efficiency

Since CTTA is in an online setting, both adaptation time and memory are critical factors. As shown in Tab. 5, TiME introduces acceptable, marginal time and memory overhead compared with the source model and Tent, while remaining substantially more efficient than other baselines. This efficiency stems from As-COOT, the core theoretical foundation of TiME, whose optimization can be efficiently solved via GPU parallelism, enabling fast optimization of the adaptation process (see complexity analysis in App. G). TiME leverages an As-COOT-driven discarding strategy to adapt on a small set of high-quality samples, reducing memory consumption. In contrast, teacher-student-based methods such as CoTTA, DSS, and CTta-T require maintaining multiple models, leading to higher time and memory. SAR, SoTTA, and REM rely on complex parameter adaptation or sample processing strategies, which increase both time and memory.

Table 5. Evaluation of time and memory efficiency of different CTTA methods on Text Understanding and Medical Reasoning task streams across Qwen and DeepSeek MoE models. Time denotes per-sample adaptation plus inference time (seconds), and Memory denotes average GPU memory usage (GB). Source denotes zero-shot inference without adaptation.

Method	Text Understanding				Medical Reasoning	
	Qwen		DeepSeek		Qwen	
	Time (s) ↓	Memory (GB) ↓	Time (s) ↓	Memory (GB) ↓	Time (s) ↓	Memory (GB) ↓
Source	3.52	29.76	3.17	30.30	3.03	30.83
Tent	3.71	30.28	4.67	32.09	3.12	31.17
CoTTA	15.68	60.21	12.46	73.37	16.63	63.28
SAR	4.35	35.84	10.85	43.09	7.03	41.95
SoTTA	6.69	37.64	10.34	45.18	5.30	42.82
DSS	10.63	59.44	11.87	73.04	11.82	60.24
REM	10.17	50.58	10.70	61.37	14.69	52.04
CTTA-T	4.32	58.92	10.14	72.41	12.67	59.27
TiME (EOE)	3.94	30.48	5.89	34.65	3.23	31.24
TiME (OR)	3.83	30.20	5.44	34.06	3.18	30.99

J. Evaluation of Time and Memory Efficiency of Each Module

We conduct a module-wise evaluation of time and memory efficiency, as shown in Tab. 6, to analyze the overhead introduced by each module in TiME.

Using expert output entropy (EOE) in TiME results in higher memory usage and time compared to original MoE routing (OR). This is because EOE requires forwarding each sample through all experts and computing per-expert output entropy, which introduces additional overhead. In contrast, OR directly leverages the pretrained router to identify compatible experts, avoiding exhaustive expert evaluation and leading to a more efficient SSA implementation. Consequently, OR achieves a better efficiency–performance trade-off and is adopted as the default strategy.

Replacing As-COOT with COOT leads to higher memory consumption. This is mainly attributed to the sample-side relaxation in As-COOT. By allowing the model to discard high-cost noisy samples, As-COOT effectively reduces the number of samples participating in adaptation, resulting in lower memory. In contrast, COOT enforces stricter mass conservation, forcing all samples to remain involved in routing and allocation, which increases memory. Notably, As-COOT achieves this efficiency improvement while also delivering superior routing quality. The variant without As-COOT exhibits higher memory usage than TiME, confirming that the TiME-based allocation mechanism itself is not only beneficial for performance but also helps control memory growth during adaptation.

Table 6. Evaluation of time and memory efficiency of each module. "r." denotes replacing the As-COOT-based allocation strategy in TiME with the corresponding alternative strategy. "w/o." denotes the removal of the module or strategy entirely.

Method	Time (s) ↓	Memory (G) ↓
Source	3.52	29.76
TiME (EOE)	3.94	30.48
TiME (OR)	3.83	30.20
r. COOT-based	4.04	31.81
w/o. As-COOT	3.67	31.75
w/o. SSA	3.75	30.07
w/o. Random Restoration	3.76	30.10

Removing SSA (§ 3.2) slightly reduces memory usage but leads to increased runtime (3.75 s vs. 3.83 s in TiME) and, as shown in the ablation study, degrades performance. This is because SSA provides semantic-level alignment between samples and experts, which helps As-COOT construct a more informative cost matrix and accelerates the convergence of the As-COOT optimization process. Without SSA, As-COOT requires more iterations to reach a stable allocation, resulting in longer runtime. Overall, SSA introduces only a small memory overhead while improving both routing quality and optimization efficiency.

Disabling random restoration (§ 3.3) results in nearly identical memory (30.10 G) and runtime (3.76 s) compared to TiME, demonstrating that re-injecting source-domain knowledge incurs almost no additional overhead. Despite its small overhead, this strategy significantly improves robustness and stability during CTTA.

K. Analyses of the Task Stream Order

We analyze the impact of the task stream order on model performance by evaluating our method and baselines under both Forward and Reverse settings across Text Understanding (TU) and Medical Reasoning (Med) task streams. As shown in the Tab. 7, we observe two distinct phenomena regarding the order of the stream:

(1) The order of the task streams noticeably impacts the stability of baseline methods, whereas our TiME remains consistent. Baselines exhibit performance volatility when the order is reversed. For instance, on the TU task stream in Qwen, REM drops from 0.4301 (Forward) to 0.4116 (Reverse), indicating that the adaptation quality of these methods depends on the specific task stream order. Conversely, our TiME demonstrates insensitivity to order permutations. Taking the DeepSeek model on TU as an example, TiME (OR) yields virtually identical results between the Forward (0.4481) and Reverse (0.4488) settings. Even on Qwen, the fluctuation is minimal compared to the clear drops seen in baselines. (2) Interestingly, on the

Table 7. Performance of different methods under different task stream orders (forward and reverse), averaged over all tasks within each task stream. In the main experiments (as shown in Tab. 1 and Tab. 4), tasks are arranged in the **Forward** order, and the **Reverse** order is constructed by reversing the forward task order. Experiments are conducted on two task streams (Text Understanding and Medical Reasoning) across Qwen and DeepSeek MoE models. Source denotes zero-shot performance without adaptation.

Method	Text Understanding				Medical Reasoning	
	Qwen		DeepSeek		Qwen	
	Forward	Reverse	Forward	Reverse	Forward	Reverse
Source	0.4449	0.4449	0.4256	0.4256	0.3576	0.3576
Tent	0.3887	0.3866	0.3537	0.3501	0.3445	0.3523
CoTTA	0.4006	0.4071	0.4106	0.4031	0.3455	0.3449
SAR	0.4234	0.4077	0.4251	0.4149	0.3522	0.3561
SoTTA	0.4261	0.4092	0.4275	0.4098	0.3592	0.3639
DSS	0.4236	0.4096	0.4103	0.4058	0.3588	0.3626
REM	0.4301	0.4116	0.4197	0.4133	0.3396	0.3379
CTTA-T	0.4548	0.4456	0.4300	0.4274	0.3607	0.3671
TiME (EOE)	0.4661	0.4616	0.4450	0.4462	0.3754	0.3808
TiME (OR)	0.4707	0.4650	0.4481	0.4488	0.3759	0.3791

TU in Qwen, we observe a universal performance decline for all methods in the Reverse order. Even the strong baseline CTTA-T drops from 0.4548 to 0.4456, and our TiME (EOE) decreases slightly from 0.4661 to 0.4616. This suggests that when the model initially adapts to a task stream that begins with tasks containing higher uncertainty or noise (as in the Reverse setting for this dataset), its adaptation capability for subsequent tasks may be compromised. Conversely, on the Med, the Reverse order often yields comparable or slightly better results (e.g., our TiME (OR) improves to 0.3791), indicating the optimal order varies by domain characteristics.

These results imply that the sequential order significantly challenges the model’s adaptation ability during CTTA. The universal drop in the Qwen-TU-Reverse setting suggests that early exposure to an unfavorable task can lead to suboptimal model adaptation. However, despite these challenging settings, our method consistently outperforms baselines in both order (e.g., surpassing the Source baseline of 0.4449 significantly even in the harder Reverse setting). This confirms that we have provided an effective online MoE routing method. By mitigating the routing errors issue inherent in original MoE models, our approach ensures the model can adapt effectively, even when the specific arrangement of the task stream imposes significant adaptation difficulties.

L. Comparison with Methods of Online MoE Re-Routing Setting

To further assess the generality and positioning of our TiME in the CTTA setting, we additionally compare TiME with several representative online MoE re-routing baselines.

Task Definition: *Online MoE re-routing* aims to address the performance degradation caused by suboptimal routing decisions when pretrained MoE routers encounter out-of-domain test samples. Since routing functions are learned on training domains, they may fail to generalize to domain shifts at test time, leading to inefficient expert selection. To this end, online MoE re-routing optimizes expert selection during testing by adapting routing behaviors based on test-time signals, while keeping the base model parameters frozen.

We note that these methods are not designed for CTTA and do not explicitly account for online domain shifts in evolving test streams, which are central to CTTA. Instead, they primarily focus on improving MoE inference by optimizing routing decisions at test time under a fixed task domain. Nevertheless, since these methods also aim to mitigate suboptimal expert selection at test time without retraining model parameters, they provide a meaningful point of comparison. Including them allows us to better contextualize our method within the broader landscape of test-time MoE routing techniques and to demonstrate the advantages of CTTA-aware routing under online and non-stationary scenarios.

We compare with the following representative online MoE re-routing methods: *THOR* (Zuo et al., 2022) randomly allocates experts to each input while enforcing routing consistency. *Rewiring* (Su et al., 2025) performs data-free self-supervision by applying lightweight updates to router logits, rewiring expert selection during testing. *R2-T2* (Li et al., 2025) conducts local optimization guided by reference samples constructed by training domain samples, enabling dynamic expert re-routing without additional training during testing.

Table 8. Comparing our TiME (CTTA setting) with different methods of *online MoE re-routing* setting on Text Understanding task stream using Gwv1.5-MoE-A2.7B-Chat. **Bold** and Underline denote the best and second-best results, respectively. *Source* denotes zero-shot performance without adaptation.

Setting	Method	t								Avg $\uparrow$
		Winogrande Acc	ARC-c Acc	ARC-e Acc	BoolQ Acc	OBQA Acc	RTE Acc	XQUAD-Zh EM	XQUAD-Es EM	
Zero-Shot	Source	0.6531	0.3179	0.6520	0.7204	0.2734	0.6458	0.2111	0.0851	0.4449
Online MoE Re-Routing	THOR	0.6421	0.3055	0.6381	0.6987	0.2692	0.6597	0.1975	0.0840	0.4369
	Rewiring	0.6656	0.3428	0.6715	0.7118	0.2780	0.6944	0.2218	0.0504	0.4545
	R2-T2	0.6712	0.3428	0.6763	0.7219	0.2795	0.7454	0.2361	0.0747	0.4685
Continual Test-Time Adaptation	TiME (ours)	0.6734	0.3368	0.6659	0.7261	0.2812	0.7361	0.2437	0.1025	0.4707

As shown in Tab. 8, our TiME consistently achieves the best average performance, showing its effectiveness in addressing test-time routing errors under CTTA. Among the compared baselines, THOR performs the worst and even underperforms the zero-shot source model on most tasks. This is expected, as THOR relies on random expert allocation with only routing consistency enforced, without exploiting any semantic or domain-aware test-time signals, making it ineffective under domain shifts. Rewiring yields only marginal gains over the source model and remains inferior to TiME. This can be attributed to its reliance on noisy contextual signals as self-supervision during unlabeled testing, as well as the lack of explicit expert load balancing, which leads to expert collapse and degrades generalization. R2-T2 is the strongest baseline among online MoE re-routing methods, achieving performance close to TiME. Its effectiveness mainly stems from retrieval-based guidance *using labeled training samples* for test-time re-routing. However, this reliance on labeled source-domain samples limits its practicality under continuous domain shifts. In contrast, TiME strictly operates in the CTTA setting without any labeled test-time samples, better reflecting real-world deployment scenarios with unlabeled and evolving test streams.

Overall, this comparison shows that while online MoE re-routing methods are effective within their own problem settings, where test-time routing is optimized for fixed or task-specific domains, their assumptions differ fundamentally from those of CTTA. In contrast, CTTA explicitly models realistic deployment conditions, characterized by evolving test domains and the absence of labeled supervision. By addressing this more practical and challenging setting, TiME is better aligned with real-world scenarios that require continuous, unsupervised adaptation to non-stationary testing domain streams.

M. Why Semantic Space Alignment Learned Offline Remains Valid in CTTA

Semantic Space Alignment module (SSA, § 3.2), which is learned in an offline preprocessing stage with fixed encoders, remains valid throughout the continual test-time adaptation (CTTA) process, although expert parameters evolve over time, and that static alignment is not only sufficient but also desirable in CTTA, and that updating SSA online may instead introduce instability and semantic drift.

First, SSA aligns sample-expert functional compatibility rather than instantaneous parameter values. The alignment

captures the relative suitability between samples and experts at the level of functional roles and specialization. Although expert parameters are updated during CTTA, these updates refine experts locally to better fit evolving domains and do not fundamentally alter their functional semantics. Therefore, the global compatibility structure learned by SSA remains stable.

Second, expert adaptation in CTTA is smooth and constrained. Updates are performed with small optimization steps and regularization (e.g., entropy minimization and random restoration), which prevents abrupt semantic shifts or role collapse. As a result, expert evolution preserves the semantic geometry in which SSA operates, ensuring that distance-based allocation costs remain meaningful over time.

Third, keeping SSA fixed is a deliberate design choice. Allowing SSA to adapt online with unlabeled and continuously shifting test data would couple alignment learning with noisy updates, potentially causing semantic drift and unstable allocation costs. A static SSA instead serves as a stable semantic anchor, enabling consistent comparison between evolving experts and current-domain samples.

Finally, even if minor misalignment arises as experts evolve, the proposed As-COOT formulation is robust to imperfect cost estimation. By relaxing the sample-side constraint, As-COOT can down-weight or discard high-cost samples, preventing residual alignment errors from contaminating routing and adaptation. Therefore, SSA only needs to provide a coarse but stable compatibility signal, which is sufficient for effective test-time routing under CTTA.

N. Additional Experiments

We further compare TiME with several CTTA baselines on the Text Understanding task stream, including BECoTTA (Lee et al., 2024a), DPCCore (Zhang et al., 2025b), and TLM (Hu et al., 2025). The results show that TiME consistently achieves stronger overall performance across diverse tasks.

Table 9. Experimental results of TiME and other CTTA baselines on the Text Understanding task stream.

Method	Winogrande	ARC-c	ARC-e	BoolQ	OBQA	RTE	XQuAD-Zh	XQuAD-Es	Avg $\uparrow$
BECoTTA	0.6346	0.3311	0.5711	0.6940	0.2773	0.6892	0.1034	0.0529	0.4192
DPCCore	0.6435	0.3267	0.6569	0.7204	0.2793	0.7071	0.1756	0.0899	0.4499
TLM	0.6531	0.3190	0.6597	0.7212	0.2793	0.6501	0.1571	0.0722	0.4390
TiME (ours)	0.6734	0.3368	0.6659	0.7261	0.2812	0.7361	0.2437	0.1025	0.4707

As shown in Tab. 9, TiME achieves the best average performance among all compared methods and consistently outperforms existing CTTA baselines.

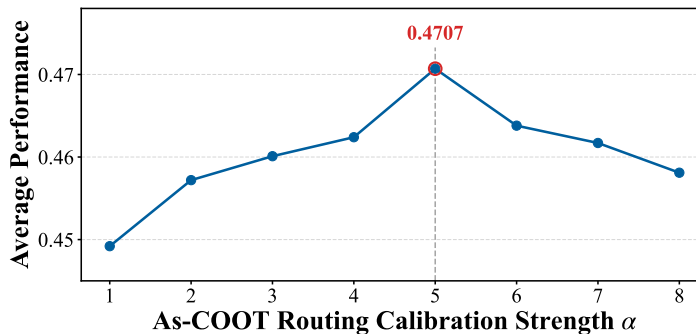


Figure 6. The model performance under different As-COOT calibration strength α .

O. Hyperparameter Analyses

O.1. Hyperparameter Analysis of As-COOT Calibration Strength α

We conduct a hyperparameter analysis on the As-COOT calibration strength α to evaluate the influence of the domain-level allocation scheme on the final routing decisions (as shown in Fig. 6). The results demonstrate that the average performance across TU tasks stream steadily improves as α increases, reaching a peak of 0.4708 at $\alpha = 5$, after which it exhibits

a declining trend. This phenomenon indicates that while a moderate α effectively incorporates real-time domain priors to correct the outdated static router, an excessively large α overshadows the essential fine-grained token-level semantic information, leading to suboptimal routing decisions.

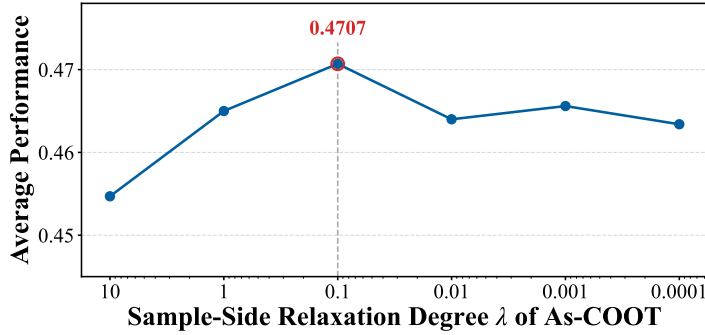


Figure 7. The model performance under different sample-side relaxation degree λ .

O.2. Hyperparameter Analysis of Sample-Side Relaxation Degree λ

We investigate the impact of the sample-side relaxation degree hyperparameter λ in As-COOT, which controls the strictness of the marginal constraint and thus the proportion of discarded samples. As shown in Fig. 7, the model performance initially improves as λ decreases from 10, reaching a peak accuracy of 0.4707 at $\lambda = 0.1$. However, further reducing λ leads to a slight performance decline. This trade-off indicates that a larger λ forces the routing of noisy samples, negatively affecting online model adaptation, while an excessively small λ results in the discarding of informative samples, hindering the model’s ability to learn from the current domain.

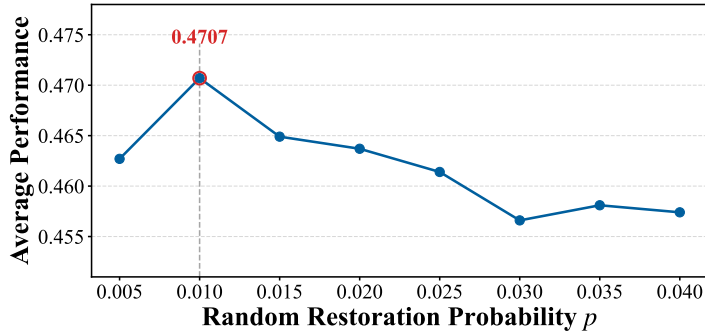


Figure 8. The model performance under different random restoration probability p .

O.3. Hyperparameter Analysis of Random Restoration Probability p

We conduct a hyperparameter analysis on the random restoration probability p to investigate its effect on balancing current domain adaptation and source knowledge preservation (as shown in Fig. 8). The experimental results show that the model performance across TU tasks stream follows a parabolic trend as p increases, achieving the optimal average performance of 0.4708 at $p = 0.01$, while further increasing p leads to a gradual decline in accuracy. This phenomenon suggests that a meticulously selected restoration probability effectively injects source domain knowledge to prevent catastrophic forgetting, whereas an excessively high p over-regularizes the model, hindering its ability to sufficiently adapt to the rapidly shifting test-time domain distributions.

O.4. Hyperparameter Analysis of Activated Expert Number K

We investigate the impact of the number of activated experts k on both adaptation performance and computational overhead (i.e., per-sample adaptation plus inference time). As shown in Fig. 9, the performance improves significantly from $k = 1$ to $k = 2$ and peaks at $k = 3$, whereas larger values ($k > 3$) lead to performance degradation and monotonically increasing

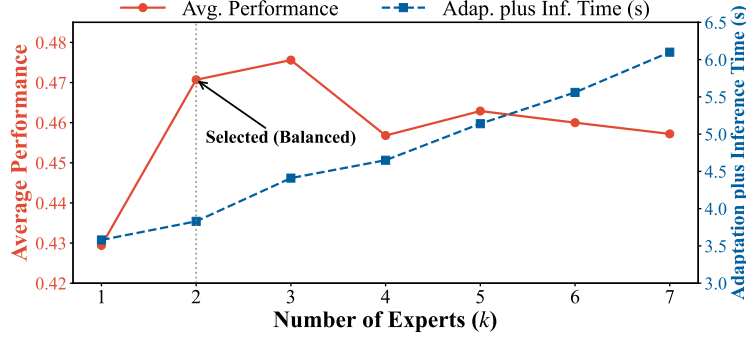


Figure 9. The MoE model performance under different Numbers of Experts Activated (k).

latency. We select $k = 2$ as the optimal configuration to balance effectiveness and efficiency. The performance drop at larger k values suggests that forcing the activation of less relevant experts introduces functional noise during the CTTA process. Since our As-COOT routing and adaptation rely on precise sample-expert compatibility, including incompatible experts dilutes the prediction accuracy and destabilizes the online parameter updates with noisy gradients.

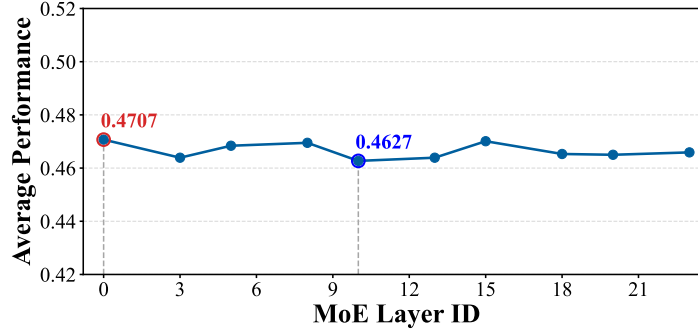


Figure 10. The model performance under different MoE layer depths.

O.5. Analysis of TiME of Different MoE layer Depths

We conduct a Hyperparameter analysis to investigate the impact of deploying our proposed TiME at different MoE layer depths. As shown in Fig. 10, results across various MoE layers in the Qwen1.5-MoE model demonstrate that the average performance on text understanding tasks stream remains consistently high and stable, with accuracy fluctuating minimally between 0.4627 and 0.4707. These results indicate that our proposed TiME is robust to the choice of layer depth, proving that the As-COOT mechanism can effectively calibrate routing and facilitate model adaptation regardless of whether it is applied to lower-level feature extraction or higher-level semantic processing. Consequently, our approach reduces the burden of hyperparameter tuning for layer selection in practical CTTA deployments.

P. Full Version of Related Work

P.1. Routing in Mixture-of-Experts LLMs

MoE routing allocates input tokens to a subset of experts. It falls into two types (Cai et al., 2025): 1) Training-based routing adopts a parameterized router to map token representations to expert weights. MoE-Multi (Shazeer et al., 2017) learns routing decisions jointly with experts via backpropagation. GShard (Lepikhin et al., 2021) trains routing end-to-end with experts while enforcing load balancing. Qwen1.5-MoE (Team, 2024) stabilizes MoE training through load balancing and entropy regularization. DeepSeek-V2 (Liu et al., 2024a) improves expert specialization by fine-grained segmentation. ReMoE (Wang et al., 2025b) proposes a differentiable ReLU-based router. Lory (Zhong et al., 2024) introduces a causal differentiable routing that retains sequence causality. MoR (Zhang et al., 2025a) enhances robustness by multiple sub-routers.

Rewiring (Su et al., 2025) leverages data-free test-time self-supervision via lightweight router logit updates to dynamically rewire MoE expert selection on-the-fly. 2) Training-free routing uses algorithmic allocation to route the expert. Hash Layer (Roller et al., 2021) uses a random router by hashing the token. THOR (Zuo et al., 2022) randomly allocates experts to each input with consistency. DEMix (Gururangan et al., 2022) allocates experts to pretraining domains and blends outputs at inference. PanGu- Σ (Ren et al., 2023) routes tokens to the expert group and randomly selects one. Symbolic-MoE (Chen et al., 2025b) routes queries to experts via inferred skill names in natural language, enabling gradient-free, adaptive reasoning. R2-T2 (Li et al., 2025) leverages test-time local optimization guided by reference samples, performing dynamic expert re-routing without additional training. Training-based routing is tightly coupled to the training domain and thus fails under domain shift during testing. Training-free routing lacks semantic modeling. We aim to design an online routing to address both limitations.

P.2. Continual Test-Time Adaptation

CTTA is a testing process that mitigates continual domain shift (Wang et al., 2022). It falls into: 1) Non-backpropagation-based methods update inputs or outputs during testing. LAME (Boudiaf et al., 2022) adjusts probabilities with Laplace maximum likelihood. FOA (Niu et al., 2024) uses covariance adaptation to learn input prompts. 2) Backpropagation-based methods update the model by optimizing using test samples. Tent (Wang et al., 2021) adapts during testing by minimizing entropy in the BN layer. EATA (Niu et al., 2022) minimizes the samples with low entropy. OIL (Ye et al., 2022) uses a teacher-student framework and causal inference to improve model adaptability. CoTTA (Wang et al., 2022) uses a teacher-student framework, data augmentation, and stochastic restoration. VIDA (Liu et al., 2023) enhances CoTTA by adding learnable lower-order and higher-order parameters. SAR (Niu et al., 2023) conducts an entropy filter and sharpness-aware entropy minimization. SoTTA (Gong et al., 2023) buffers class-balanced samples and sharpens entropy minimization. BECoTTA (Lee et al., 2024a) leverages mixture-of-domain low-rank experts with input-dependent domain-adaptive sparse routing. DeYo (Lee et al., 2024b) adapts image data with highly consistent predictions before and after disruption. DSS (Wang et al., 2024b) mitigates noise by prediction confidence for positive and negative learning. PCoTTA (Jiang et al., 2024) focuses on point cloud understanding. MoTTA (Zhou et al., 2025) filters noise via model-pruning and uses flatness-aware minimization. NF-CTTA (Rafi et al., 2025) selects a high-confidence subset based on entropy and neighbor similarity. TLM (Hu et al., 2025) proposes test-time learning, a TTA-like paradigm that adapts by minimizing input perplexity. DPCore (Zhang et al., 2025b) uses visual prompt adaptation and prompt coreset preservation. REM (Han et al., 2025) builds the prediction difficulty via a progressive masking strategy. CTTA-T (Liu et al., 2025) perceives domain shift and refines noise via prediction consistency. Existing work focuses on the dense model. Our goal is to explore a CTTA method for Mixture-of-Experts (MoE), as the sparsity of the MoE architecture is more suitable for CTTA than dense.

Table 10. Statistical significance (p -values) of performance improvements of our TiME on Text Understanding (TU) and Medical Reasoning (Med) task streams across different MoE models. All values are obtained from paired t -tests.

Text Understanding (TU)									
Model	Method	Winogrande	ARC-c	ARC-e	BoolQ	OBQA	RTE	XQUAD-Zh	XQUAD-Es
Qwen	TiME (EOE)	1.09E-05	0.0065	2.76E-09	0.0022	8.46E-32	6.99E-13	4.77E-05	6.99E-19
	TiME (OR)	0.0004	8.78E-13	1.63E-31	0.0014	2.08E-06	5.94E-08	0.0069	1.12E-07
Deepseek	TiME (EOE)	0.0008	0.0002	5.74E-05	2.89E-12	0.0001	7.82E-17	0.0001	1.46E-29
	TiME (OR)	8.68E-05	0.0064	0.0019	7.12E-04	9.72E-23	2.41E-11	3.18E-04	6.83E-08
Medical Reasoning (Med)									
Model	Method	ICD9-CM	ICD10-CM	ATC	ICD9-PROC	ICD10-PROC	MedQA	MedMCQA	PubMedQA
Qwen	TiME (EOE)	9.48E-07	3.68E-16	0.0007	1.94E-33	0.0004	5.74E-51	4.94E-11	3.66E-07
	TiME (OR)	3.72E-12	1.84E-03	1.76E-05	6.11E-24	1.23E-08	2.46E-39	7.58E-06	0.0028

Q. Significance Test

To examine the statistical significance of the performance gains brought by our approach, we conduct t -tests (Bartlett, 1937) on the primary experimental results. All reported results are obtained by averaging over five independent runs with different random seeds. As shown in Tab. 10, under both the TU and Med task streams, the performance improvements of our method over the baseline models are statistically significant, with all p -values below 0.01.